Blackhole Tracker: BHSM Boundary Interpretation of Compact-Lens Mass, Motion, and Conditional Response

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Abstract

Blackhole Tracker provides reproducible lens inference with explicit Berger–Hopf Standard Model (BHSM) interpretation: one effective exterior is counted once. A local app, command line, Python package and notebooks support resolved image-pair data and native unresolved-photometry likelihood. An internally frozen pilot processes 100 OGLE-III events and 76 344 measurements. Eighty-nine fail the specified photometric adequacy screen; eleven are not rejected; two pass stricter archival follow-up criteria. Both are published events. Their timescales do not identify mass, distance or absolute motion; no new black-hole interpretation is established. A 2022 benchmark rejects rectilinear photometry-only fits. A separately versioned joint benchmark uses 12 521 photometric measurements and 21 two-coordinate HST positions. Three frozen noise candidates converge but fail residual-structure checks; their masses remain conditional. Earlier resolved-pair synthetic calibrations remain separately versioned. Reviewed corrections remove numerical-reference dependence, admit physical nuisance profiles and preserve unavailable motion uncertainties. The optional scalar–tensor calculation bounds boundary states in its retained exterior, but two projected modes vanish and the leading apparent-mass response is degenerate with shear. The software supports auditable event screening and conditional response tests. Neither BHSM core dynamics, a detectable envelope, nor a horizon classification is measured by these observations.

Keywords: strong gravitational lensing; compact objects; microlensing; native photometry; scalar–tensor gravity; boundary response; identifiability

1 Introduction: the Blackhole Tracker question

Lensing already measures the mass and motion of dark compact objects. Astrometric microlensing has been used to infer an isolated stellar-mass black-hole candidate's properties [1]; joint astrometry and parallax separate familiar mass, distance and proper-motion combinations. Moving dark structures also motivate multi-source astrometric templates [2], and repeated lensed bursts carry motion-dependent delays [3]. The product makes this measurement workflow reproducible under an explicit data contract. Its action-specific question is narrower: can a specified envelope deformation supply a calculable linked pattern in images, fluxes and delays, and which parts survive an ordinary lens fit?

BHSM organizes its research around boundary geometry, constrained actions and envelope matching. That language does not by itself fix a metric, surface state or optical coefficient. An ordinary gravitational potential may already describe the same boundary contribution. We therefore use a declared Einstein/GHY sector for the ordinary fit and a single scalar–tensor action

for the optional R1 response, counting each physical contribution once. The name *Blackhole Tracker* denotes the proposed application to a candidate compact dark source. All rays in this paper lie far outside a declared worldtube, itself far outside the gravitational radius. Lensing on that domain does not establish a horizon or identify the compact interior.

The real-observation extension adds a separate unresolved-photometry route and freezes its selection, likelihood and all-event reporting before fitting. It preserves the downloaded measurements instead of forcing them into a resolved-image contract. Its observational outcome is event screening with mass unresolved, including a retained negative benchmark result; it does not manufacture a black-hole classification.

The central deliverable is a usable research product with an ordinary compact-lens tracking mode and an optional conditional boundary-response mode. The ordinary fit is interpreted in the recovered BHSM Einstein/GHY sector through the same exterior mass and its boundary representation; this interpretation does not add another gravitational contribution. The new theoretical result is a uniform sufficient admission bound on the normalized quadrupole throughout the retained R1 exterior. The optical calculation identifies exact blind directions and demonstrates why a working compact-mass track need not detect an envelope. A separate nonlinear product validation tests mass/motion recovery, full covariance, limited interval calibration and unsuccessful controls. These statements are independent of accepting a departure from GR; their action-specific application remains conditional.

1.1 Recovered record and scope of the new calculation

A finite recovery covered relevant local Git branches/worktrees, research files in Downloads and Documents, selected members of a mounted BHSM backup, and accessible excerpts of the conversation *Strong Lensing Paper Plan*. The source map and content hashes are distributed with the calculation. This is actual access to selected material, not comprehensive recall of every conversation or archive.

The earlier angular paper was recovered from `BHSM_Angular_Lensing_Reproducibility.zip`, SHA256 beginning `84786b6b905ae1e2`. Its round closed-background, nine-harmonic, six-coordinate calculation is preserved separately. Its illustrative fixed-physical-source SIS delay shift, -0.353216090 s against a 7.753621449 d baseline, is a conditional chosen state, not a selected black-hole signal. The earlier observational HE0435 paper and its failed or incomplete observational tests are also preserved. No such data are used here. A separate Hopf-charge result is retained as supporting research rather than made the central tracker claim.

The strongest feasible action-specific deformation route is thus a normalized *conditional effective-action* calculation. We do not substitute a familiar surface mass identity for the missing horizon problem. The spherical flux is kept as an exterior matching datum; the physical state remains free. The mathematical chain, and the place where it stops, are made explicit.

2 BHSM interpretation of the reconstructed exterior

The ordinary lens fit is part of the BHSM interpretation used here. It reconstructs the effective gravitational exterior of the compact configuration: its monopole, location, translation and admitted projected structure. An observable departure from GR is not required for that interpretation. The recovered BHSM source records explicitly admit an Einstein/GHY compact-source sector, in which boundary and conventional gravitational descriptions represent the same exterior contribution. We use that declared sector without promoting it into a unique microscopic derivation of every fitted parameter.

Let $\mathcal{C}$ denote a configuration described in the BHSM framework, and let $W_{\mathcal{C}} = \Phi_{\mathcal{C}} + \Psi_{\mathcal{C}}$ be its

Table 1: Recovered objects relevant to the tracker. "Not located" refers only to the searched record. Exact paths/hashes are in the source map.

Object	Scientific status and use
R1 covariant action and homogeneous background	Conditional effective realization; archived source and hashes recovered and replayed. Supplies the local coefficients used here.
Compact-source cubic reduction	Recovered action-derived quasi-static constraints and regular-body branch. Re-derived and checked here with general exterior scalar flux.
BHSM black-hole/surface drafts	Surface identities and proposed flux parameters recovered. Horizon selection of the R1 exterior flux and causal deformation response not established by those objects.
Microscopic constrained Hessian/Schur work	Actual local response machinery recovered; an evaluated normalized microscopic-to-cosmological optical map was not located.
Fixed external environment	Seven action-owned interface variations remain even when external labels are fixed. This microscopic prescription does not fix a cosmological or black-hole deformation state.

Table 2: Interpretation dictionary for the ordinary optical fit. All entries remain conditional on the supplied geometry and finite lens model.

Fitted quantity	BHSM interpretation and what is constrained
Optical mass $M_{\rm opt}$	Effective gravitational monopole of the represented configuration. It is the point-exterior lens coefficient, not a measurement of a microscopic constituent mass or a surface energy without its matching relation.
Reference center $\boldsymbol{c}_0$	Location of that optical configuration in the supplied tangent-plane frame.
Observer proper motion $\boldsymbol{\mu}_{\rm obs}$	Translation of that same configuration per observer year. It does not measure internal envelope evolution or select a causal relaxation law.
Compact quadrupole	Projected nonspherical structure of the same exterior; an envelope may be represented by this term when it produces that potential. The coefficient does not uniquely identify its cause.
Shear and convergence	The admitted projected environmental response. Their interpretation within a BHSM environment does not assign them uniquely to the tracked compact object.

Weyl potential in the stated weak-metric convention. The forward map is

$$\mathcal{C} \longmapsto W_{\mathcal{C}} \longmapsto \psi_{\mathcal{C}} \longmapsto (\boldsymbol{\theta}_+, \boldsymbol{\theta}_-, \ln R, \Delta t), \tag{1}$$

with the supplied distances, clock, angular frame and source positions. When a source convention writes $W_{\rm topo} = (\Phi_{\rm topo} + \Psi_{\rm topo})/2$, its contribution to our full Weyl potential is $2W_{\rm topo}$; this conversion creates no second physical response. The fit constrains the projected response represented by $\psi_{\mathcal{C}}$. Two descriptions giving the same projected potential on the relevant image neighborhoods give the same positions, magnification determinants and relative Fermat delays. Their likelihood and fitted track are consequently identical. This is a physical interpretation of the reconstructed exterior, rather than a demand for an additional force or a second mass contribution.

The spherical boundary interpretation can be made explicit without changing the lens equations. In the static Schwarzschild member of the declared Einstein sector, flat-reference Brown–York energy obeys

$$E_{\rm BY}(r) = \frac{c^4 r}{G}\left(1 - \sqrt{1 - \frac{2GM_{\rm opt}}{c^2 r}}\right), \qquad M_{\rm opt} = \frac{E_{\rm BY}}{c^2} - \frac{GE_{\rm BY}^2}{2c^6 r}, \quad r > \frac{2GM_{\rm opt}}{c^2}. \tag{2}$$

This is the established gravitational boundary construction [4], also recovered in the BHSM source record. The corresponding principal extrinsic-curvature deficit of the enclosing sphere, relative to

its counterpart in a flat spatial slice, and its inverse mass relation are

$$\delta k_g(r) = \frac{1-\sqrt{1-2GM_{\rm opt}/(c^2 r)}}{r}, \qquad M_{\rm opt} = \frac{c^2}{G}\left(r^2\delta k_g - \frac{1}{2}r^3\delta k_g^2\right). \tag{3}$$

Indeed the two principal curvatures of the enclosing sphere in the static spatial slice are $\sqrt{f}/r$, compared with $1/r$ in the flat reference. This is not its intrinsic Gaussian curvature, which is $1/r^2$ in both cases at the same areal radius. For fixed declared r, let $f = 1 - 2GM_{\rm opt}/(c^2 r)$. Local propagation of the fitted monopole uncertainty gives $\sigma_{E_{\rm BY}/c^2} = \sigma_M/\sqrt{f}$ and $\sigma_{\delta k_g} = G\sigma_M/(c^2 r^2\sqrt{f})$. These are derived representations of the same fitted coefficient, not independent measurements or a fit of the enclosing radius. If software uses a computational default for r, that choice must remain explicit. The errors are first-order fixed-radius pushforwards, not finite Gaussian intervals when a mass range reaches the static-branch boundary $f = 0$. The radius-dependent boundary energy and the conserved monopole are representations of the same spherical solution. Their contributions are counted once. The fit does not measure a new worldtube radius, and changing the enclosing radius in this identity is not physical mass drift. This static local example does not assert a global cosmological embedding or continue a static boundary observer through a horizon.

The conditional R1 extension supplies a more specific response relation from its adopted action, frozen background and supplied matching data. Its scalar-driven Weyl response may be added only as a distinct term relative to the already represented reference configuration; an envelope already absorbed into an ordinary metric multipole cannot be added again under another name. If two parameterizations produce the same potential, their degeneracy must be retained. In an R1 fit the reported mass is the action's metric/source mass parameter; its relation to a point-exterior $M_{\rm opt}$ must include the scalar contribution to the total potential. An ordinary quadrupole is not automatically a normalized R1 boundary amplitude. The forward transfer, state convention and nuisance freedoms must be specified before that further interpretation is made.

Accordingly a supported mass/motion track is a reconstruction of the effective BHSM-described optical configuration in the declared sector. It need not distinguish that interpretation from another theory giving the same exterior. It does not separately infer the microscopic state, an astrophysical horizon, scalar flux selection or deformation dynamics. Those additional inferences require their corresponding source and matching relations; they are not conditions for retaining the ordinary fit or for using its mass, location and motion within the stated BHSM interpretation.

Core and surface: the dated interpretation amendment. Norman P. Carberry's 22 October 2026 proposal distinguishes a common nongeometric core, described by abstract mode weights and algebraic phase relations, from spacetime-side fields and interfaces [5]. No metric, clock, trajectory, volume or propagation speed is assigned inside that core. A harmonic label acquires a physical frequency or length only through a specified realization; the surface S^3 or Berger–Hopf Laplacian is not thereby a core Laplacian. The existing regular, same-spacetime enclosure carrier and reciprocal attachment equations retain their own domains. They are not relabeled as a terminal transfer into the core.

The amendment distinguishes propagating excitations, operative-interface envelopes and persistent boundary-dominated encapsulations. The inherited differential is primary in the proposed account; formation, maintenance and terminal transition are separate questions. Sustained black-hole conversion, a distinct quiet neutron-star endpoint, quantum loss/regain of surface manifestation and historical galaxy boundary persistence are attributed hypotheses, not additional findings of this lens fit. In particular a proposed terminal channel with zero spacetime daughters still requires charge/flux accounting and a physical matching law. The enclosing radius in Eqs. (2)–(3) is not a measurement of the proposed hard interface. No new optical term or core parameter follows from this interpretive update.

Reviewed radial mechanics and recovered source versions. The corrected amendment [5] makes the radial membrane reduction's domain explicit. For a round spatial p-sphere on a fixed Minkowski background, constant tension γ, $c = 1$, coordinate-time derivatives and $|\dot{R}| < 1$, take

$$L_{\mathrm{mem}} = -\gamma \Omega_p R^p \sqrt{1 - \dot{R}^2}, \qquad L_{\mathrm{work}} = \frac{\Delta P\, \Omega_p R^{p+1}}{p+1}. \tag{4}$$

Here the second term represents a constant applied normal pressure jump; variation gives

$$\gamma \left[\frac{\ddot{R}}{(1-\dot{R}^2)^{3/2}} + \frac{p}{R\sqrt{1-\dot{R}^2}} \right] = \Delta P. \tag{5}$$

The pure membrane term gives zero traction, not arbitrary pressure forcing. A dynamic pressure source must supply its actual variation. For a regular static metric $ds^2 = -f\,dt^2 + dr^2/f + r^2 d\Omega_p^2$ the outward static curvature is $K = f'/(2\sqrt{f}) + p\sqrt{f}/R$, reducing to p/R only for $f = 1$. Curved/backreacting applications need their actual induced metric and traction equation, and varying tension supplies additional variation terms. Spherical symmetry alone does not justify Eq. (5).

The source reconciliation preserves the implementation-era handoff whose hash begins `d604c8c4`, including its two blank filename bullets, separately from the named handoff `149ba5ef` and the recovered original front and galaxy-envelope PDFs `7bb85093` and `de228c96`. All four raw hashes verify; both PDFs were read and their 23 pages inspected. Their mathematical statements agree with the amendment at their declared levels: the front calculation is a prescribed-background scalar restriction, and the galaxy persistence/radial-source forms are conditional ansatz results. They supply neither a closed gravitational interface nor selected physical coefficients, lifetime, source export or core optical transfer. The source ledger therefore records recovered evidence without silently replacing historical bytes or expanding a restricted derivation into a full coupled solution. Twenty-two symbolic consequence checks and ten focused source/mechanics tests pass at formal theory revision `705d4c7e`.

What a further core-to-optical calculation must determine. There is a useful conditional response criterion, beyond merely naming a matching relation. Let $\boldsymbol{a}$ be a declared differentiable chart of abstract mode data, with its admissible topology specified; it is distinct from an existing regular child-state coordinate and from the prescribed R1 boundary amplitudes a_A used elsewhere in this paper. Let Q collect all regular metric, field and interface data entering the observable vector $\boldsymbol{o}$, together with the constraints and gauge choice. If a physical correspondence specifies $F_\Sigma(Q, \boldsymbol{a}) = 0$ and $K = D_Q F_\Sigma$ has an appropriate bounded inverse on the declared constrained physical tangent, differentiation gives

$$D_{\boldsymbol{a}} Q = -K^{-1} D_{\boldsymbol{a}} F_\Sigma, \qquad D_{\boldsymbol{a}} \boldsymbol{o} = -D_Q \boldsymbol{o}\, K^{-1} D_{\boldsymbol{a}} F_\Sigma. \tag{6}$$

This form assumes that Q includes every optical input apart from supplied geometry. If an independently relevant input is omitted, its explicit $D_{\boldsymbol{a}}\boldsymbol{o}|_Q$ term must also be retained. For the existing linear regular-side constrained response $Ky = f$, the corresponding operator-derivative identity is $D_{\boldsymbol{a}} y = K^{-1}[(D_{\boldsymbol{a}} f) - (D_{\boldsymbol{a}} K)y]$, with the declared retarded child recovery. It does not replace an unknown core derivative by a regular child jet.

For the full supplied covariance $C = LL^{\mathsf{T}}$ and whitened nuisance Jacobian N, the locally retained response is

$$J_{\mathrm{core}}^{\mathrm{prof}} = (I - NN^{+}) L^{-1} D_{\boldsymbol{a}} \boldsymbol{o}, \qquad F_{\mathrm{core}} = (J_{\mathrm{core}}^{\mathrm{prof}})^{\mathsf{T}} J_{\mathrm{core}}^{\mathrm{prof}}. \tag{7}$$

The pseudoinverse uses the declared local nuisance rank. Only directions outside the kernel of this profiled map can be distinguished. Optical projection nulls cannot be removed by renaming a

regular mode a core harmonic. These equations specify a conditional calculation; the physical chart/domain and $D_{\boldsymbol{a}}F_\Sigma$ still have to come from the core-to-regular correspondence. An unavailable transfer is not a zero response, and the present software reports no numerical core information matrix, transition rate or daughter probability.

Prospective discovery use. The engine is intended to help identify previously uncatalogued compact lenses and black-hole candidates from supplied observations. Its useful output is a supported effective mass, location and motion when the supplied data identify them, under the declared BHSM interpretation and without requiring a non-GR residual. The separate native-photometry pilot in Section 10 instead reports timescale/blending constraints and keeps mass, distance and absolute motion unavailable. Catalogue novelty is an association/cross-match statement. Distinguishing a black hole from an unresolved stellar system, multiple lens or other compact interior additionally requires suitable observations and model comparison. The supplied distances, parity association, geometric flux and delay calibration, source freedoms, environment, noncritical image branches and local identifiability must remain adequate. This release does not acquire sky images or claim a newly observed object.

3 Source action, exterior boundary response and optical normalization

This section defines the precise conditional exterior problem used by the tracker. The action, background and source pairing are supplied; a physical black-hole horizon relation and the microscopic BHSM-to-R1 identification are not inferred. The worldtube below is outside the unresolved compact interior and is not a horizon boundary condition.

3.1 Common action and local source

In signature $(-+++)$, initially setting $c=\hbar=1$ and $M_{\rm Pl}^{-2}=8\pi G$, the retained action is

$$S=\int\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R-\frac{1}{2}(\nabla T)^2+\frac{(\nabla T)^2\Box T}{\lambda}-V_0+\mu^3 T\right]d^4x+S_m. \tag{8}$$

Here physical $\lambda=\Lambda^3$, while the pinned numerical input is $\widehat{\lambda}=\lambda/(M_{\rm Pl}H_0^2)=1$. This is a cubic kinetic-braiding action in the established scalar–tensor setting [6, 7]. With $\Pi_{\mu\nu}=\nabla_\mu\nabla_\nu T$, variation gives

$$\Box T+\frac{2}{\lambda}\left[\Pi_{\mu\nu}\Pi^{\mu\nu}-(\Box T)^2+R_{\mu\nu}\nabla^\mu T\nabla^\nu T\right]+\mu^3=0. \tag{9}$$

It reproduces the frozen homogeneous current equation $J=v+6Hv^2/\lambda$, $\dot{J}+3HJ=\mu^3$, where $v=\dot{\bar{T}}$. Thus the local source sign follows from the same action and background.

For a minimally coupled point-matter model, $S_{\rm pp}=-M\int ds$ and $\delta S_{\rm pp}=\frac{1}{2}\int\sqrt{-g}T^{\mu\nu}_{\rm pp}\delta g_{\mu\nu}d^4x$. In its nonrelativistic local limit the excess density is $M\delta^{(3)}(\boldsymbol{r})$, with no direct scalar insertion $\delta S_{\rm pp}/\delta T$. Strongly self-gravitating horizon response does not follow from treating that point-matter action as a black hole.

Subtract the homogeneous equations, write $T=\bar{T}(t)+\pi_T(r)$ and freeze v, $d=\ddot{\bar{T}}$ and H during a local encounter. At the retained weak-metric, quasi-static order, keeping the nonlinear scalar

Hessian terms, the scalar and metric rows are

$$A\Delta\pi_T - \frac{2}{\lambda}[(\Delta\pi_T)^2 - |\operatorname{Hess}\pi_T|^2] + b\Delta\Phi = 0,$$
$$2M_{\rm Pl}^2\Delta\Phi = \delta\rho - b\Delta\pi_T, \qquad \Phi = \Psi,$$
$$b = \frac{2v^2}{\lambda}, \qquad A = 1 + \frac{4(d+2Hv)}{\lambda}. \tag{10}$$

The density is the local excess, not the homogeneous cosmic density counted again. Eliminating that same metric row gives

$$C\Delta\pi_T - \frac{2}{\lambda}[(\Delta\pi_T)^2 - |\operatorname{Hess}\pi_T|^2] = -\frac{b\,\delta\rho}{2M_{\rm Pl}^2}, \qquad C = A - \frac{b^2}{2M_{\rm Pl}^2}. \tag{11}$$

For example, varying the static functional

$$S_{\rm QS} = \int dt\, d^3r \left[-\frac{C}{2}|\nabla\pi_T|^2 + \frac{|\nabla\pi_T|^2\Delta\pi_T}{\lambda} + \frac{b\,\delta\rho\,\pi_T}{2M_{\rm Pl}^2}\right] \tag{12}$$

reproduces Eq. (11). Its last term is the consumed source pairing after metric elimination, not an additional covariant matter coupling. Higher weak-field orders need not retain $\Phi = \Psi$.

3.2 Spherical exterior, scales and open compact-object flux

Let $L_0 = c/H_0$, $x = r/L_0$, $\widehat{m} = (GM/c^2)/L_0$, $\widehat{v} = v/(M_{\rm Pl}H_0)$, and $\widehat{d} = d/(M_{\rm Pl}H_0^2)$. Define

$$\widehat{\beta} = \frac{b}{M_{\rm Pl}} = \frac{2\widehat{v}^2}{\widehat{\lambda}}, \quad C = 1 + \frac{4(\widehat{d} + 2\widehat{H}\widehat{v})}{\widehat{\lambda}} - \frac{\widehat{\beta}^2}{2}, \qquad p(x) = \frac{d(\pi_T/M_{\rm Pl})}{dx}. \tag{13}$$

Regular-origin matching for an ordinary finite matter source gives the exterior flux $-\widehat{\beta}\widehat{m}$. To expose the uncompleted compact-object matching, retain a supplied nonnegative flux ratio χ:

$$Cx^2p - \frac{4}{\widehat{\lambda}}xp^2 = -\chi\widehat{\beta}\widehat{m},$$
$$p(x) = \frac{-2\chi\widehat{\beta}\widehat{m}}{x^2\left[C + \sqrt{C^2 + 16\chi\widehat{\beta}\widehat{m}/(\widehat{\lambda}x^3)}\right]},$$
$$\Phi_x = \Psi_x = \frac{\widehat{m}}{x^2} - \frac{\widehat{\beta}}{2}p(x). \tag{14}$$

This root is continuous with the decaying linear response. The metric monopole mass remains M; χ changes the scalar flux, not the mass. The value $\chi = 1$ is the regular-matter reference, not a derived black-hole charge. A horizon-regular, cosmologically matched solution is required to determine the physical horizon relation. A negative flux is outside the real, positive-discriminant branch implemented here.

The exterior screening scale and extra force fraction are

$$r_V^3 = \frac{16\chi\widehat{\beta}\,(GM/c^2)L_0^2}{\widehat{\lambda}C^2}, \quad \epsilon_\infty = \frac{\chi\widehat{\beta}^2}{2C}, \quad F(r) = \frac{\delta\Phi'}{(GM/c^2)/r^2} = \frac{2\epsilon_\infty}{1 + \sqrt{1 + (r_V/r)^3}}. \tag{15}$$

The scalar field is not the provider's time-displacement column: $\pi_T = -v\pi_{\rm time}$ when reconstructing it from uniform-field gauge. Adding it directly as another gravitational potential would change both its normalization and the source equation.

At $z_l = 0.3$ the pinned background replay gives

$$\widehat{\beta} = 0.113063844895134, \quad C = 4.23885855943370, \quad D_t = 4.39622102870814, \tag{16}$$

where $D_t = 1 + 12\widehat{H}\widehat{v}/\widehat{\lambda} + 6\widehat{v}^4/\widehat{\lambda}^2$ is the homogeneous leading time principal coefficient. For $M = 10^9 M_\odot$, $H_0 = 67.4$ km/s/Mpc, $\chi = 1$ and $r_0 = 10$ pc, $(L_0, GM/c^2, r_V) = (4.4479593175 \times 10^9, 4.7854158416 \times 10^{-5}, 45680.3117591)$ pc. The compactness at the worldtube is 4.7854×10^{-6} and $H_l r_V/c = 1.2156 \times 10^{-5}$. These are local expansion parameters, not measured error bars. Near a horizon the weak-metric approximation fails; the worldtube must remain outside that domain.

3.3 Exact linear multipoles and canonical boundary response

Write the dimensionless scalar perturbation as $\delta\widehat{\pi} = a f_\ell(r) Y_{\ell m}$, with $\int |Y_{\ell m}|^2 d\Omega = 1$. Linearizing Eq. (11) gives

$$\delta E = Z^{ij}\partial_i\partial_j\delta\pi = \partial_i(Z^{ij}\partial_j\delta\pi), \qquad Z^{ij} = C\delta^{ij} - \frac{4}{\lambda}[(\Delta\pi_0)\delta^{ij} - \Pi_0^{ij}], \tag{17}$$

because $\partial_i Z^{ij} = 0$. The anisotropic Hessian term is essential. The spherical eigenvalues satisfy

$$K_r = C - \frac{8\pi_0'}{\lambda r}, \quad K_\Omega = C - \frac{4}{\lambda}(\pi_0'' + \pi_0'/r), \quad K_r' = \frac{2(K_\Omega - K_r)}{r}. \tag{18}$$

Outside the source the normalized radial equation is therefore

$$\frac{1}{r^2}(r^2 K_r f_\ell')' - \frac{\ell(\ell+1)K_\Omega}{r^2} f_\ell = 0. \tag{19}$$

An actual density multipole adds the fixed source $-b\delta\rho_{\ell m}/(2M_{\rm Pl}^2)$ to the right side and requires interior matching. A prescribed boundary value is not that interior solve.

For $\rho = r/r_V$ and $u = \rho^{-3}$, the exact exterior coefficients are

$$K_r = C\sqrt{1+u}, \quad K_\Omega = C\frac{1+u/4}{\sqrt{1+u}}, \quad K_t = D_t + \frac{3(S-C)^2}{4S} \geq D_t > 0, \quad S = C\sqrt{1+u}. \tag{20}$$

Thus this retained scalar problem is elliptic and has a positive leading time coefficient. This is not full-system stability or a horizon/UV certificate; its scalar characteristic speeds can exceed the metric light speed in part of the exterior.

The decaying solution of Eq. (19) is

$$R_\ell(\rho) = \rho^{-\ell-1}{}_2F_1\left(\frac{\ell+1}{2}, \frac{\ell+2}{6}; \frac{2\ell+4}{3}; -\rho^{-3}\right), \qquad f_\ell(r) = \frac{R_\ell(r/r_V)}{R_\ell(r_0/r_V)}. \tag{21}$$

It follows by substituting $t = -\rho^{-3}$ in the radial equation. The deep-screened powers are $(\ell+1)/2$ and $-\ell/2$; the far powers are ℓ and $-\ell-1$. In particular the decaying quadrupole has r^{-1} and r^{-3} limits. For $\ell = 1$, ${}_2F_1(1, 1/2; 2; -u) = 2/(1+\sqrt{1+u})$, so f_1 is exactly proportional to $-p$: translating the monopole reproduces the linear dipole equation. This supplies an independent field-equation check. The $\chi = 0$ limit is $f_\ell = (r_0/r)^{\ell+1}$.

The exterior quadratic energy, divided by its overall $M_{\rm Pl}^2 L_0$ factor, is

$$\frac{1}{2}\int_{x_0}^{\infty}[x^2 K_r(\partial_x\delta\widehat{\pi})^2 + \ell(\ell+1)K_\Omega\delta\widehat{\pi}^2]dx = \frac{1}{2}\Lambda_\ell a^2, \qquad \Lambda_\ell = -x_0^2 K_r(x_0) f_{\ell,x}(x_0) > 0. \tag{22}$$

The inner normal of the exterior domain is $-\partial_r$; its conjugate flux is $q_{\ell m} = \Lambda_\ell a$. Positivity establishes uniqueness of the zero-data decaying exterior solution. Holding a fixed and holding $q_{\ell m}$ fixed are different source comparisons. Neither amplitude is selected by the action without an interior/boundary state. The numerical reference has $\Lambda_2 = 0.00294226449484$ in these dimensionless-x conventions.

The same metric constraint yields $\delta\Phi = \delta\Psi = \delta\Phi_N - \widehat{\beta}\delta\widehat{\pi}/2 + H_{\rm match}$, where $\Delta H_{\rm match} = 0$. Ordinary gravitational multipoles and outer matching remain independent until a common source fixes them. The implemented scalar-driven term adds no arbitrary metric quadrupole.

3.4 Projected phase, derivatives and finite matching

For a transverse real quadrupole choose

$$Y_N = \sqrt{\frac{15}{16\pi}}(n_x^2 - n_y^2) \equiv N_2(n_x^2 - n_y^2), \qquad W \equiv \Phi + \Psi = -\widehat{\beta} a f_2(r) Y_N. \tag{23}$$

For impact $b \geq r_0$, every integration point lies outside the worldtube. The scalar projected physical phase length is

$$\begin{aligned} \delta\widehat{\psi}(b,\varphi) &= a\, q(b) \cos 2\varphi, \\ q(b) &= -\widehat{\beta} N_2 \int_{-\infty}^{\infty} f_2(\sqrt{b^2+z^2}) \frac{b^2}{b^2+z^2}\, dz \\ &= -2\widehat{\beta} N_2 b \int_0^{\pi/2} f_2(b\sec\theta) d\theta. \end{aligned} \tag{24}$$

Both halves of the line are included. With $r = b\sec\theta$,

$$\begin{aligned} q'(b) &= -2\widehat{\beta} N_2 \int_0^{\pi/2} (f_2 + r f_2') d\theta, \\ q''(b) &= -\frac{2\widehat{\beta} N_2}{b} \int_0^{\pi/2} (r^2 f_2'' + 2 r f_2') d\theta. \end{aligned} \tag{25}$$

The provider returns q, q', q'' in pc, dimensionless units and pc^{-1}, respectively. Exact hypergeometric connection formulae preserve the cancellation of the deep r^{-1} contribution in these radial derivatives. Projection gives a deep phase plateau and a far b^{-2} phase, rather than identifying a local force ratio with a projected photon deflection.

An optional axial boundary coordinate uses the independently normalized $Y_{20} = N_0(3n_z^2 - 1)$, $N_0 = \sqrt{5/(16\pi)}$. Its physical phase per unit axial boundary amplitude is

$$q_0(b) = -2\widehat{\beta} N_0 b \int_0^{\pi/2} (2\sec^2\theta - 3) f_2(b\sec\theta) d\theta. \tag{26}$$

Its first and second radial derivatives replace f_2 by $f_2 + r f_2'$ and $r^2 f_2'' + 2 r f_2'$, respectively, with prefactors $-2\widehat{\beta} N_0$ and $-2\widehat{\beta} N_0/b$. This supplies a separately normalized radial nuisance when the axial state is not fixed. It is distinct from varying the spherical scalar flux. Its leading deep-screened phase is logarithmic and can be mass-like. In the exact zero-flux harmonic limit its full-line projection vanishes; the transverse phase is then exactly $q = -4\widehat{\beta} N_2 r_0^3/(3b^2)$. The implementation uses these analytic zero-flux identities instead of an interpolation residual.

The scalar monopole deflection is supplied separately:

$$\delta\widehat{\alpha}_0(b) = -\frac{\widehat{\beta} b}{L_0} \int_{-\infty}^{\infty} \frac{p(r/L_0)}{r}\, dz = \frac{4GM}{c^2 b} \int_0^{\pi/2} F(b\sec\theta)\cos\theta\, d\theta. \tag{27}$$

The ordinary $4GM/(c^2 b)$ is counted once by the lens model. A relative spherical phase is obtained by integrating this deflection between impacts; its formally divergent absolute line potential is unnecessary.

Infinity in these expressions is a matched thin-lens extension, not an exact static spacetime. A local overlap cutoff obeys $\max(b, r_V) \ll L \ll c/H_l, R_{\mathrm{curvature}}$. The scalar monopole tail, divided by $4GM/(c^2 b)$, is bounded at leading order by $\epsilon_\infty b^2/(2L^2)$. If $A_2 = r_V^3/R_2(r_0/r_V)$, the Euler integral of the hypergeometric function gives $f_2(r) \leq A_2/r^3$ and hence

$$|q_{|z|>L}| \leq \frac{\widehat{\beta} N_2 A_2 b^2}{2L^4}. \tag{28}$$

These control the stated tail only. They do not bound coefficient time dependence, a changed environment, or a different physical outer state.

3.5 Perturbative example and actual numerical control

For the supplied state convention $a = \eta x_0 |p(x_0)|/6$, $\eta = 10^{-3}$, the numerical reference has $a = 1.37809214552 \times 10^{-16}$. The closed R1 distances at $(z_l, z_s) = (0.3, 1)$ give an Einstein impact 339.621287478 pc and $aq(r_E)/(4GM/c^2) = -8.891115935 \times 10^{-13}$. These amplitudes are declared inputs, not horizon-selected values.

On 65 radii from r_0 to $10r_V$ and 100 directions, the sampled maximum perturbation/background scalar-Hessian norm ratio is 3.742×10^{-4}. The exact linearized equation residual, scaled by its principal Hessian terms, is below 6.1×10^{-15}. The positive comparison scale used here is

$$\mathcal{S} = \left[C + \frac{4}{\lambda}(|\operatorname{tr}\Pi_0| + \|\Pi_0\|_F) \right] \|\delta\Pi\|_F. \tag{29}$$

The retained nonlinear scalar remainder is exactly $-2[(\operatorname{tr}\delta\Pi)^2 - \delta\Pi : \delta\Pi]/\lambda$; its sampled ratio to the stated principal-term scale is at most 8.57×10^{-5}. These are finite-sample diagnostics, not uniform interval certificates.

Independent radial integration agrees with the special function to 2.25×10^{-10} relative over the tested domain, with the outer decaying condition imposed at $1000r_V$. The exact translation check is within 1.23×10^{-15}. Independent original-line quadrature checks the projected quadrupole phase to 2.23×10^{-16} pc and the scalar monopole deflection to 4.04×10^{-27} at $b = 1000$ pc. The code records Gauss-order refinement and Hermite interpolation discrepancies for every prepared table. Its fast potential, gradient and Hessian derive from one interpolating potential, preserving their mutual consistency.

Slow source motion requires small velocity relative to the scalar characteristic speeds; a varying multipole additionally requires a quasi-static crossing condition such as $\omega r_V / c_{\text{scalar}} \ll 1$. The retained field equations omit higher weak-metric and time-dependent terms. A nonzero environmental scalar Hessian modifies the same operator and cannot generally be added by linear superposition. The paper's homogeneous-background exterior avoids that extra assumption. The positive principal coefficients, small illustrative amplitude and tiny formal delay do not establish physical precision at that delay, an observational forecast, or a solved black-hole boundary problem.

4 A uniform admission bound for boundary states

The exterior action gives a sufficient amplitude domain before any optical fit. Let $\mathcal{H}_0 = \operatorname{Hess}_x \widehat{\pi}_0$ and $\delta\mathcal{H} = \operatorname{Hess}_x \delta\widehat{\pi}$, with derivatives in $x = r/L_0$. In an orthonormal real quadrupole basis, write

$$\delta\widehat{\pi} = f_2 \sum_{A=1}^{5} a_A Y_A = N_{\text{STF}} f_2 \, n^i \mathsf{Q}_{ij} n^j, \qquad N_{\text{STF}} = \sqrt{\frac{15}{8\pi}}, \quad \mathcal{A} = \|\boldsymbol{a}\|_2 = \|\mathsf{Q}\|_F, \tag{30}$$

where Q is symmetric and trace free and $f_2(x_0) = 1$. The result covers all five coordinates, including the two erased by a symmetric line-of-sight projection. A smaller template family specifies its omitted coordinates separately. The product uses the two transverse and one axial coordinates, with the two odd coordinates fixed to zero; their absence is a model convention, not a measurement of their values.

Proposition 1 (Uniform retained-exterior control)**.** *Assume the isolated decaying branch, $\chi > 0$, $P_0 > 0$, and $C, D_t, \widehat{\lambda} > 0$. Put $P_0 = x_0|p(x_0)|$, $B = 12N_{\text{STF}}$, $d_* = D_t/C$, and*

$$R_t = \begin{cases} 6, & d_* \geq 3/4, \\ 6t(2-t), \quad t = \sqrt{4d_*/3}, & 0 < d_* < 3/4. \end{cases} \tag{31}$$

At every $x \geq x_0$ and every angular direction,

$$\frac{\|\delta\mathcal{H}\|_F}{\|\mathcal{H}_0\|_F} \leq \frac{B\mathcal{A}}{\sqrt{2}P_0}, \qquad \frac{\|\delta Z\|_{\mathrm{op}}}{K_\Omega} \leq \frac{2\sqrt{2}B\mathcal{A}}{P_0},$$
$$\frac{|\delta K_t|}{K_t} \leq \frac{4\sqrt{3}B\mathcal{A}}{R_t P_0}. \tag{32}$$

Here $\delta Z = -4[(\operatorname{tr}\delta\mathcal{H})I - \delta\mathcal{H}]/\widehat{\lambda}$ and $\delta K_t = -4\operatorname{tr}\delta\mathcal{H}/\widehat{\lambda}$ are the retained principal perturbations.

Proof. Set $k = K_\Omega/K_r \in [1/4, 1]$ and $y = d\log f_2/d\log r$. Equation (19) gives

$$r^2 f_2'' + 2kr f_2' - 6k f_2 = 0, \qquad \frac{dy}{d\log r} = (1-2k)y + 6k - y^2. \tag{33}$$

At $y = -3$ and -1 the outward vector fields are respectively $12(k-1) \leq 0$ and $8k - 2 \geq 0$. Thus $[-3, -1]$ is invariant under inward integration and contains the decaying far-field branch $y = -3 + 9(r_V/r)^3/8 + O((r_V/r)^6)$. Hence $-3 \leq y \leq -1$. The elliptic maximum principle gives positive f_2 and also

$$f_2(x) \leq f_1(x) = \frac{|p(x)|}{|p(x_0)|}. \tag{34}$$

Indeed $L_2 f_1 = -4K_\Omega f_1/r^2 < 0$, whereas $L_2 f_2 = 0$; their normalized inner and decaying outer boundary data agree. Here $L_\ell f = r^{-2}(r^2 K_r f')' - \ell(\ell+1)K_\Omega f/r^2$.

Rotate the evaluation direction to the third axis. The Hessian map on the orthogonal $m = 2, 1, 0$ matrix subspaces has singular coefficients

$$2, \qquad 2|y-1|, \qquad \frac{2}{\sqrt{3}}\sqrt{1+2k^2}\,(3-y), \tag{35}$$

after removing $N_{\mathrm{STF}} f_2/x^2$. Each is at most 12, so $\|\delta\mathcal{H}\|_F \leq B\mathcal{A} f_2/x^2$. Now use Eq. (34), $\|\mathcal{H}_0\|_F \geq \sqrt{2}|p|/x$, $\|(\operatorname{tr} H)I - H\|_{\mathrm{op}} \leq \sqrt{2}\|H\|_F$, and $K_\Omega \geq 2|p|/(\widehat{\lambda}x)$. Finally, with $s = K_r/C \geq 1$, Eq. (20) gives

$$\frac{K_t}{|p|/(\widehat{\lambda}x)} = 6 - \frac{6}{s} + \frac{8d_*}{s-1} \geq R_t. \tag{36}$$

Taking the infimum over s yields Eq. (31); $|\operatorname{tr} H| \leq \sqrt{3}\|H\|_F$ completes the result. □

For declared budgets $\epsilon_H > 0$, $0 < \epsilon_Z < 1$ and $\eta_{\max} > 0$, a sufficient state cap is

$$\mathcal{A} \leq \mathcal{A}_{\max} = P_0 \min\left\{\frac{\eta_{\max}}{6}, \frac{\sqrt{2}\epsilon_H}{B}, \frac{\epsilon_Z}{2\sqrt{2}B}, \frac{\epsilon_Z R_t}{4\sqrt{3}B}\right\}, \qquad \eta = \frac{6\mathcal{A}}{P_0}. \tag{37}$$

This preserves the retained principal signs throughout the exterior and bounds the relative Hessian. These are approximation budgets, rather than a selected astrophysical amplitude. The cap is sufficient, not necessary: exceeding it is outside this guarantee and the product's declared admitted family; it does not establish physical impossibility or an actual change of principal sign. The exact quadratic scalar remainder obeys $|\mathcal{R}_2| \leq 4\|\delta\mathcal{H}\|_F^2/\widehat{\lambda}$; a residual bound alone is not an inverse-operator error estimate. The conormal map is exact for the linear exterior:

$$\boldsymbol{q} = \Lambda_2 \boldsymbol{a}, \qquad \boldsymbol{a} = \boldsymbol{q}/\Lambda_2, \qquad \frac{E_2}{M_{\mathrm{Pl}}^2 L_0} = \frac{1}{2}\Lambda_2 \mathcal{A}^2. \tag{38}$$

Thus specified conormal data are admitted when $\|\boldsymbol{q}\|_2 \leq \Lambda_2 \mathcal{A}_{\max}$. This source pairing does not select a horizon flux or interior state.

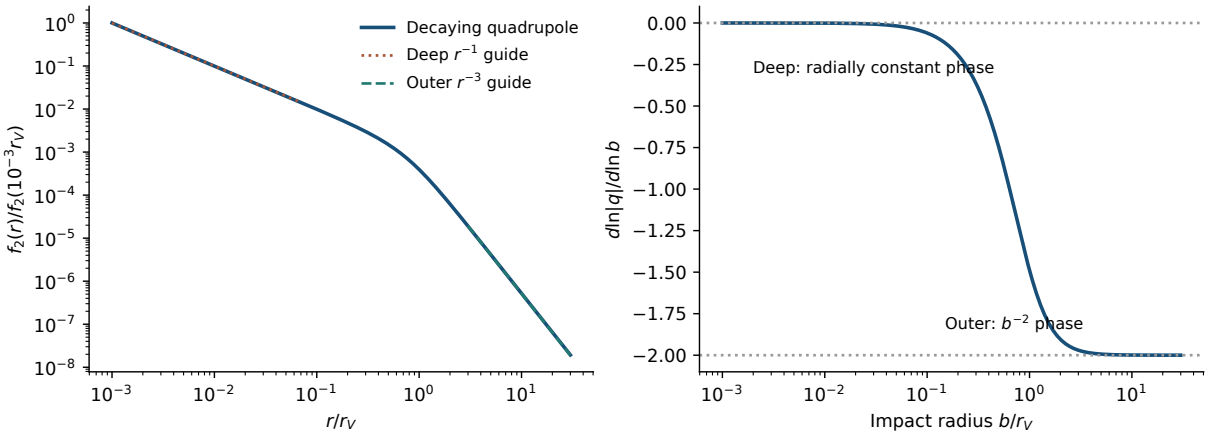


Figure 1: Conditional exterior transfer for the supplied regular-body flux proxy. Left: normalized decaying quadrupole, with its known deep r^{-1} and outer r^{-3} powers. Right: the slope of the fully matched projected phase, approaching zero and -2. The curves use the action-fixed coefficients and the full line integral, not an extrapolation of either power law across the transition.

At $\chi = 0$, the relative normalization P_0 vanishes but nonzero boundary states remain possible. Here $f_2 = (x_0/x)^3$, $K_r = K_\Omega = C$, $K_t = D_t$, and $\Lambda_2 = 3Cx_0$. The absolute cap

$$\mathcal{A} \le \epsilon_Z \widehat{\lambda} x_0^2 \min\left\{\frac{C}{4\sqrt{2}B}, \frac{D_t}{4\sqrt{3}B}\right\} \tag{39}$$

preserves the same retained principal signs; relative η is undefined.

Admission also requires weak metric potentials, rays with $b \ge r_0$, and a local patch whose background changes slowly. A useful uniform spatial-gradient control is $[(|p_0| + \sqrt{6}N_{\rm STF}\mathcal{A}/x_0)/|\widehat{v}|]^2 \ll 1$. For a patch scale L, the implemented drift estimate is $\max(|\widehat{H}|, 2|\widehat{d}/\widehat{v}|)L/L_0$. The spherical-background sound-speed lower bound $c_{\rm bg}$ follows by minimizing $\min(4s^2, s^2+3)/[3(s-1)^2 + 4d_* s]$ over $s \ge 1$. Here its minimum is $c_{\rm bg}^2$ in units of c^2. For a state with the spatial and time bounds $e_s < 1, e_t$ in Eq. (32), the Rayleigh quotient instead gives $c_{\rm min} = c_{\rm bg}\sqrt{(1-e_s)/(1+e_t)}$. The implementation uses this conservative state-inclusive bound. Require $v_\perp/(cc_{\rm min}) \ll 1$ and $L/(cc_{\rm min}\tau_{\rm local}) \ll 1$ for intrinsic evolution. Observer proper motion gives $v_\perp = |\mu_{\rm obs}|D_l(1+z_l)$; the observer timescale is $\tau_{\rm obs} = (1+z_l)\tau_{\rm local}$. Fixed boundary coefficients are an explicit time assumption. The product uses IAU nominal $GM_\odot$ for its mass-length conversion; this unit convention leaves the equations and bounds unchanged. These controls apply to the retained local equations. Cosmological matching, the coupled horizon problem and ultraviolet completion require their own calculations.

5 Projection and the information retained by the tracker

The screened multipole exponents and hypergeometric radial modes have cubic-Galileon precedents [8–10]. Here the input is the normalized exterior response of the preceding section, with its fixed scalar-to-metric coefficient. The statements below concern that conditional response, rather than a horizon-selected amplitude.

5.1 A normalized STF boundary state and its projected nulls

Let C be a real symmetric trace-free 3×3 matrix, normalized by $\operatorname{tr}\mathsf{C}^2 = 1$, and define

$$Y_{\mathsf{C}}(\mathbf{n}) = N_{\rm STF} C_{ij} n_i n_j, \qquad N_{\rm STF} = \sqrt{\frac{15}{8\pi}}, \qquad \int_{S^2} Y_{\mathsf{C}}^2 \, d\Omega = 1. \tag{40}$$

Thus $af_2(r)Y_{\mathsf{C}}$ has boundary L^2 norm $|a|$ at r_0. For $\mathsf{C}_+ = \mathrm{diag}(1,-1,0)/\sqrt{2}$, this is precisely the transverse harmonic with $N_2 = \sqrt{15/(16\pi)}$ used above; $N_{\mathrm{STF}}/\sqrt{2} = N_2$. Define $\mathcal{W} = \Phi + \Psi$. Its scalar-driven response is $\delta\mathcal{W} = \mathcal{A}_W f_2 Y_{\mathsf{C}}$, with $\mathcal{A}_W = -\widehat{\beta}a$ for π_T/M_{Pl} normalization. This is the fixed action coefficient, not an additional photon coupling.

For a frozen straight ray with $\mathbf{x} = (b\cos\varphi, b\sin\varphi, z)$, $b \geq r_0$, factor out $N_{\mathrm{STF}}\mathcal{A}_W$ and the known geometric prefactor. The remaining full-line projection is

$$\mathcal{P}_f[\mathsf{C}](b,\varphi) = \int_{-\infty}^{\infty} f(r)\frac{C_{ij}x_i x_j}{r^2}\,dz, \qquad r^2 = b^2 + z^2. \tag{41}$$

Parameterize the five STF components by

$$\mathsf{C} = \begin{pmatrix} -c/2 + s_1 & s_2 & u \\ s_2 & -c/2 - s_1 & v \\ u & v & c \end{pmatrix}, \qquad \mathrm{tr}\,\mathsf{C}^2 = \frac{3}{2}c^2 + 2(s_1^2 + s_2^2 + u^2 + v^2). \tag{42}$$

Direct integration gives

$$\mathcal{P}_f[\mathsf{C}] = cA_0(b) + T(b)(s_1\cos 2\varphi + s_2 \sin 2\varphi), \tag{43}$$

$$T(b) = b^2\int_{-\infty}^{\infty}\frac{f(r)}{r^2}\,dz, \qquad A_0(b) = \int_{-\infty}^{\infty} f(r)\left(1 - \frac{3b^2}{2r^2}\right)dz. \tag{44}$$

The xz and yz terms are odd in z and vanish. This exact two-dimensional kernel prevents reconstruction of an unrestricted three-dimensional STF state from this one-view projection. If its three visible coefficients are inferred with the stated action normalization, they only imply

$$a^2 \geq \frac{3}{2}(ac)^2 + 2[(as_1)^2 + (as_2)^2]. \tag{45}$$

Conditional rank-three result. For the isolated cubic response with

$$K_r = C_0\sqrt{1 + (r_V/r)^3}, \quad K_\Omega = K_r + \frac{r}{2}K_r', \quad C_0 > 0, \quad r_V > 0, \tag{46}$$

the normalized positive decaying solution obeys $T(b) > 0$ and $A_0(b) > 0$. Hence (43) has rank three as a map into projected functions, with kernel exactly (u, v). To prove the axial assertion, put $Y = r^3 f$. The radial equation implies

$$(K_r r^{-4}Y')' = 6K_r' r^{-5}Y. \tag{47}$$

Positivity of f follows from the elliptic maximum principle. Since $K_r' < 0$ and $K_r r^{-4}Y' \to 0$ at infinity, integration from r to infinity gives $Y' > 0$. An integration by parts, using $zf(r) \to 0$, then gives

$$A_0(b) = \frac{1}{2}\int_{-\infty}^{\infty}\frac{z^2}{r^2}[rf'(r) + 3f(r)]\,dz = \frac{1}{2}\int_{-\infty}^{\infty}\frac{z^2}{r^4}Y'(r)\,dz > 0. \tag{48}$$

The genuinely screened hypothesis matters: for $f = C_o/r^3$, $A_0 \equiv 0$ and $T = 4C_o/(3b^2)$, so the rank is two. This harmonic limit agrees with the transverse STF mass-multipole projection [11]. Rank in a projected representation does not by itself establish rank in lens data after monopole and source freedoms are fitted.

These full-line statements use the isolated frozen exterior as a local matching model. Finite observer/source endpoints, nonconstant geometric weights, motion during ray crossing, and cosmological outer completion require their own projection. They cannot be assumed to remove the nulls without calculation.

5.2 Radial discrimination and its limitation

In a screened overlap $r_0 \le b \ll r_V$, the matched $f \sim A_d/r$ gives

$$T(b) \sim 2A_d. \tag{49}$$

The leading transverse potential is therefore constant in radius and spin two in angle. The deep axial integral, temporarily cut off at a symmetric Z, is

$$A_0(b;Z) = A_d\left[2\,\mathrm{asinh}(Z/b) - \frac{3Z}{\sqrt{Z^2+b^2}}\right] \sim A_d[2\ln(2Z/b) - 3]. \tag{50}$$

Outer matching fixes the constant and subleading corrections. The leading $-2A_d \ln b$ is degenerate with a point-mass potential. Its additive constant has no deflection or differential-delay signal. The numerical calculation uses the matched profile rather than extending r^{-1} to infinity.

At the potential level, a constant spin-two coefficient differs from an isolated compact harmonic quadrupole proportional to b^{-2} and a uniform external shear proportional to b^2. If a calibrated spin-two potential amplitude is restricted to

$$p_2(b) = q + \mu b^{-2} + \gamma b^2, \qquad D = b\partial_b, \tag{51}$$

then

$$(D^2 - 4)p_2 = -4q. \tag{52}$$

At three distinct positive radii, writing $x_i = b_i^2$, the design rows $(1, x_i^{-1}, x_i)$ have determinant

$$-\frac{(x_2 - x_1)(x_3 - x_1)(x_3 - x_2)}{x_1 x_2 x_3} \neq 0. \tag{53}$$

Two radii cannot separate q from both nuisance coefficients. For the actual transfer the first column is $T(b_i)$, and its determinant and conditioning must be evaluated. These are conditional potential-reconstruction statements. They are not a claim that three image radii directly measure three potential values. Internal/external multipole reconstruction already has an extensive lensing literature [12, 13].

An arbitrary extended matter distribution is a stronger nuisance family: $\nabla_\perp^2[q\cos 2\varphi] = -4q\cos 2\varphi/b^2$, so an annular distributed quadrupolar density can reproduce this potential. A small quadrupolar perturbation can be supported by a positive monopole. Consequently the radial filter does not identify a gravity theory or exclude an ordinary extended envelope.

5.3 Timing blindness, scalar-estimator degeneracy, and image geometry

Use dimensionless reference potential $\psi_0 = \ln r$, source $\mathbf{y} = u\mathbf{e}$, $s = \sqrt{u^2+4}$ and primary images $\mathbf{x}_+ = a\mathbf{e}$, $\mathbf{x}_- = -d\mathbf{e}$, where $a = (u+s)/2$, $d = a^{-1}$. For a perturbation $q\cos 2(\varphi - \varphi_q)$ constant in radius, both unperturbed image potentials agree. Fermat stationarity therefore gives zero first-order delay change at fixed source. Its log flux-ratio change is

$$\delta \ln R = -4q\cos 2(\varphi_s - \varphi_q)\frac{s}{u}. \tag{54}$$

Tangential image displacements can retain the angular component absent from this scalar correction. These statements concern continued nonsingular primary branches, with perturbations small enough to avoid a critical crossing or flux-order reversal.

The first-variation operator of the point-lens apparent-mass estimator satisfies

$$\mathcal{K}_1(u) = \mathcal{K}_{r^2}(u) = \frac{s^3}{uh(u)}, \qquad h(u) = \frac{1}{2}us + 2\,\mathrm{asinh}(u/2). \tag{55}$$

Thus $(r^2-1)\cos 2\varphi$ is an exact first-order null of this scalar statistic for every $u>0$. A deep constant spin-two response and external shear cannot be separated by that statistic alone. The point-lens delay/flux relation is established prior art [14]; the evaluated transfer response, rather than different radial powers by themselves, determines its information. The unit $r^2\cos 2\varphi$ profile here has coefficient one in the potential. In the product's shear convention $\psi_{\rm shear}=\gamma_+ r^2\cos 2\varphi/2$, it corresponds to $\gamma_+=2$. The identity concerns a first variation of this scalar statistic, not an equality of finite perturbed lens models. The leading delay null likewise assumes the stated point-lens reference and fixed source; finite matching corrections to the constant radial profile can lift it.

Fixed-data gradient proposition. Let observed image positions $\boldsymbol{\theta}_\pm$ and reference Jacobians $A_\pm$ be fixed, and hold the time-delay prefactor fixed. A perturbing potential p can preserve these positions with one freely shifted source if and only if

$$\nabla p(\boldsymbol{\theta}_+)=\nabla p(\boldsymbol{\theta}_-)=-\delta\boldsymbol{\beta}. \tag{56}$$

Under (56), the remaining first-order null tests are

$$\delta\Delta\tau=(\boldsymbol{\theta}_+-\boldsymbol{\theta}_-)\cdot\delta\boldsymbol{\beta}+p_+-p_-, \tag{57}$$

$$\delta\ln R=\mathrm{tr}(A_+^{-1}Hp_+)-\mathrm{tr}(A_-^{-1}Hp_-). \tag{58}$$

The lens equations and Fermat stationarity prove these identities directly. In a forward calculation at fixed source, the images move and the flux derivative also includes the displaced background Hessian; that term cannot be dropped.

For the scalar-estimator null $p=(r^2-1)\cos 2\varphi$, direct differentiation gives

$$\nabla p(a\mathbf{e})-\nabla p(-d\mathbf{e})=2s\cos 2\varphi_s\,\mathbf{e}. \tag{59}$$

With a calibrated center, a free source alone cannot absorb this difference when $\cos 2\varphi_s\neq 0$. An added mass perturbation $\delta m\ln r$ cancels it only for $\delta m=-2\cos 2\varphi_s$, which then changes the mass estimator by δm. Thus image geometry and the estimator distinguish this particular difference if mass is the only additional compensator and the time-distance conversion is fixed. Equivalently, the geometric angular-mass product of the two radii changes by zero for constant spin two and by $2\cos 2\varphi_s$ for r^2 spin two. Comparing angular and physical masses requires distance calibration.

For $\cos 2\varphi_s=0$, the same difference is a genuine first-order joint null for a single freely positioned source: its gradients agree, its source shift is tangential, and both delay and flux-ratio variations vanish. Additional epochs with changing angle can change this rank. Moving centers, convergence, distance factors, ordinary multipoles, and calibration parameters must be included before profiling the full data Jacobian. More general source-position and multi-plane transformations impose further limits [15–18]. No unrestricted reconstruction theorem follows from the potential-space filter.

5.4 Incoming, local and outgoing optical transfer

We use the homogeneous closed R1 background for the incoming and outgoing distances. If $K=0.018$ in units $L_0=c/H_0$, define

$$\chi(z)=\int_0^z\frac{dz'}{\widehat{H}(z')},\qquad S_K(\chi)=\frac{\sin(\sqrt{K}\chi)}{\sqrt{K}},\quad D_{ab}=\frac{L_0S_K(\chi_b-\chi_a)}{1+z_b}. \tag{60}$$

The local deflection then enters the usual single-plane composition,

$$\boldsymbol{\beta}=\boldsymbol{\theta}-\frac{D_{ls}}{D_s}\widehat{\boldsymbol{\alpha}}(D_l(\boldsymbol{\theta}-\boldsymbol{c})),\qquad \widehat{\boldsymbol{\alpha}}=\nabla_b\int(\Phi+\Psi)\,dz. \tag{61}$$

This is the homogeneous limit of general optical transfer, whose equivalent-lens and source-transformation freedoms must be retained [16–18]. No angular environmental harmonic from the earlier cosmological paper is added here: a second nonlinear scalar Hessian background would change the local response operator and require a new reduction.

For a reference mass M, let $m = GM/c^2$,

$$r_E^2 = 4m\frac{D_l D_{ls}}{D_s}, \qquad \boldsymbol{x} = \frac{D_l(\boldsymbol{\theta} - \boldsymbol{c})}{r_E}, \qquad t_M = \frac{4m}{c}(1 + z_l). \tag{62}$$

The dimensionless lens potential is the projected Weyl phase divided by $4m$. In particular, the ordinary monopole is $\ln r$, $r = |x|$; the scalar monopole correction is derived from the same action. It is added to that GR metric once. If $q(b)$ is the phase in physical length per unit transverse boundary amplitude,

$$\delta\psi = \epsilon_+ Q(r)\cos 2\varphi + \epsilon_\times Q(r)\sin 2\varphi, \quad Q(r) = \frac{q(r_E r)}{q(r_E)}, \quad \epsilon_A = \frac{a_A q(r_E)}{4m}. \tag{63}$$

The a_A are supplied scalar states, not extra optical couplings. Homogeneous metric quadrupoles, which need not be fixed by a_A, are retained as ordinary compact-lens nuisance parameters.

6 A tracker statistic and its blind direction

For a point source lensed by an isolated point mass, let $R > 1$ be the absolute primary-image flux ratio. The familiar delay law [14] gives

$$u = 2\sinh(\ln R/4), \qquad s = \sqrt{u^2 + 4},$$
$$h(u) = \tfrac{1}{2}us + 2\,\mathrm{asinh}(u/2), \qquad M_{\rm app} = \frac{c^3\Delta t}{4G(1 + z_l)h(u)}. \tag{64}$$

The apparent mass is constant for any slow constant-mass point-lens trajectory, including source or observer motion incorporated in u. This baseline estimator is inherited. It also remains equal to the point mass in a constant isotropic convergence in the ideal point-lens model: rescaling the image/source coordinates leaves the dimensionless point-mass delay difference and flux relation intact. It is not a general observable invariant for arbitrary lenses.

For a small additional potential ϵp, Appendix A derives

$$\delta \ln M_{\rm app} = \frac{\delta\Delta\tau}{h(u)} - \frac{s^2}{4h(u)}\delta \ln R. \tag{65}$$

Image displacements must enter the magnification derivative even though Fermat stationarity removes them from the first-order delay at fixed source. For $p = q(r)\cos 2\varphi$, write its response as $\delta \ln M_{\rm app} = \mathcal{K}_q(u)\cos 2\varphi_s$. An exact first-order identity is

$$\mathcal{K}_1 = \mathcal{K}_{r^2} = K_*(u), \qquad \mathcal{K}_{r^{-2}} = (u^2 + 1)K_*(u), \qquad K_* = \frac{(u^2 + 4)^{3/2}}{uh(u)}. \tag{66}$$

Thus $(r^2 - 1)\cos 2\varphi$ is invisible to this statistic at first order for every $u > 0$. The potentials differ, and their images, fluxes and delays generally differ. The cancellation occurs only after compressing those data into $M_{\rm app}$. In particular, a leading screened $\epsilon\cos 2\varphi$ produces the same apparent-mass history as a conventional shear $\gamma = 2\epsilon$ of the same orientation. Different radial phase powers alone do not establish identification.

For the radially constant phase, the two primary-image angles differ by π, so $p(x_+) = p(x_-)$ and $\delta\Delta\tau = 0$. Nevertheless $\delta \ln R = -4\epsilon s\cos 2\varphi_s/u$, and tangential astrometry can also change. At $\cos 2\varphi_s = 0$ even the flux/mass statistic is blind at first order while tangential shifts remain possible. Full transition matching restores a small radial and delay response (Figure 2); this does not make the leading degeneracy irrelevant.

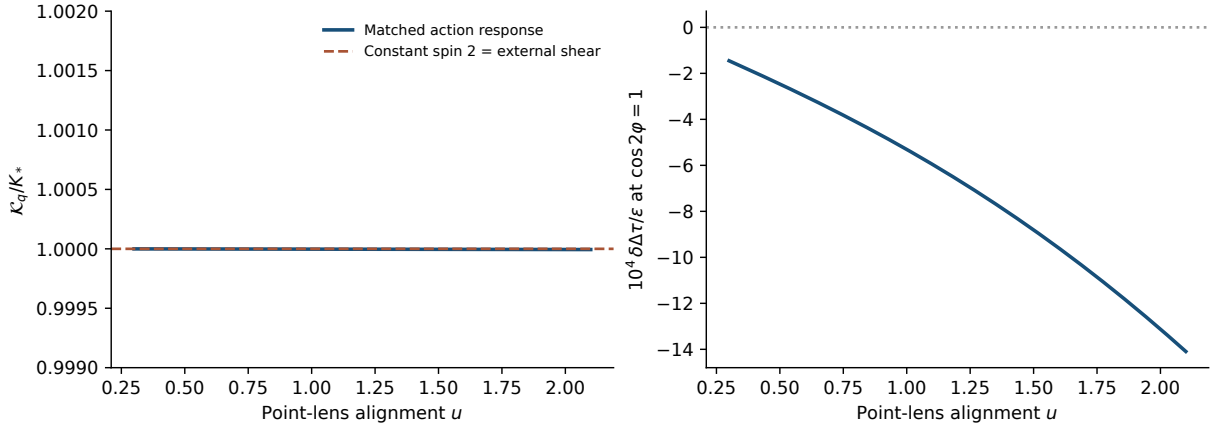


Figure 2: The matched radial profile in the point-mass first-variation operator, with $r_E/r_V = 0.00743$. Left: its apparent-mass response is close to the exactly coincident constant-phase/external-shear response, despite different potentials. Right: the small matching-induced delay derivative; the strict constant-phase result is zero. Curves are derivatives per dimensionless phase amplitude, not physical observed signals.

7 Joint primary-pair information experiment

7.1 Declared data and nuisance family

The numerical experiment uses the action-derived scalar monopole as well as the quadrupole, rather than fitting a point-mass background to an unlabeled modified profile. Its observables at each of nine reference configurations are

$$O_i = (x_{+,i}, y_{+,i}, x_{-,i}, y_{-,i}, \ln R_i, \Delta t_i/t_M). \tag{67}$$

The source reference positions are $(1.8t_i, 0.6)$, with t_i uniformly spanning $[-1, 1]$. In the fit, two source coordinates are *free at every configuration*; no known unlensed track is supplied. They can absorb parallax, source motion and source-position errors at these sample points. The lens nuisances are one mass, two center coordinates, two constant-velocity coordinates, one convergence, two external-shear and two compact material-quadrupole coordinates. The scalar flux is also fitted through its local monopole response, rather than assumed known in the inference. The visible axial scalar quadrupole is also fitted as a radial nuisance with its normalized Y_{20} response. There are 30 nuisance parameters and two transverse envelope coordinates in 54 data components. The two odd projected tensor modes are absent from this one-view data model rather than artificially fixed by a prior.

The ordinary quadrupoles are $B_A r^{-2}$ and $\Gamma_A r^2$ in the cosine/sine basis. They are fixed as a family before inspecting the envelope projection. Internal and external radial multipole information is familiar in lens modeling [13]; it is not supplied here as a new general principle. The inference assumes known redshifts, distances, time calibration and a common absolute astrometric frame. Unknown frame drift is equivalent to part of the freely fitted center motion; arbitrary frame distortion or an unrestricted radial potential is not covered. Fluxes are intrinsic geometric point-source fluxes with no extinction, blending, independent microlensing or finite-source structure. These are strong experimental assumptions, not observational facts.

For each baseline image, $A = I - D^2\psi_0$. The computed response includes

$$\delta x = A^{-1}(\nabla\delta\psi + \delta y), \qquad \delta \ln|\mu| = \mathrm{tr}\left[A^{-1}(D^2\delta\psi + D(D^2\psi_0)[\delta x])\right]. \tag{68}$$

For a translated center, $\delta\psi = -\delta c \cdot \nabla\psi_0$. Mass differentiation changes the action's scalar monopole too. Source variation enters the delay as $(x_+ - x_-)\cdot\delta y$, in addition to the potential difference.

Table 3: Declared conditional example and local information result. The last two rows use the small supplied state at source $(0.9, 0.6)$; the singular values use unit optical phase amplitudes and the stated dimensionless data metric.

Quantity	Value
Mass	$10^9 M_\odot$
Worldtube / Einstein radius	10.0 / 339.62 pc
Screening radius	45.68031 kpc
Worldtube compactness	4.79×10^{-6}
Boundary amplitude / optical amplitude	1.38×10^{-16} / -8.89×10^{-13}
Joint-pair nuisance rank	30
Projected envelope singular values	3.13473, 0.80220
Retained column-norm fractions	0.04213, 0.14512
Largest tested physical coordinate shift	9.36×10^{-5} nanoarcsec
Tested physical log-flux-ratio shift	2.88×10^{-12}

These terms ensure comparison of consistent physical states and prevent spurious source/center signatures.

Let N and S denote the nuisance and envelope derivative matrices. With a specified data covariance C_O, local information is the Gram matrix of

$$S_\perp = (I - P_N) C_O^{-1/2} S, \tag{69}$$

where P_N projects onto the column space of $C_O^{-1/2} N$. This is ordinary profiled linear inference. Here all data components have equal *dimensionless* errors in the displayed units, an explicit choice for information comparison. It is not an instrument noise model. Nuisance columns are normalized before SVD; the recorded rank threshold is 10^{-9} of the largest singular value, and the full spectra are released. Raw column norms below 10^{-11} times the larger of the maximum norm and unity are discarded as numerical zeros. All 30 full-pair columns survive that admission. The axial mode is nearly masslike; when all admitted columns have full rank but condition number exceeds 10^6, the projection uses 70-digit normal equations after the unchanged SVD rank selection. An independent 70-digit QR solve checks the result. These refine arithmetic for the same double-input derivatives, not the physical approximation. Parameter priors are not added to fill nulls.

7.2 Evaluated response and controls

For $M = 10^9 M_\odot$, $z_l = 0.3$, $z_s = 1$, $r_0 = 10$ pc and scalar flux proxy equal to the regular-body value, the same-background distances give $r_E = 339.62$ pc, $\theta_E = 74.116$ mas and $t_M = 25612.55$ s. The computed screening radius is $r_V = 45680.31$ pc. This scale is action-derived after choosing the matching flux; it is not tuned to the rays.

The joint-pair projected envelope singular values are approximately $(3.135, 0.802)$ per unit phase amplitude. Roughly 4–15% of each envelope column's norm survives the 30 nuisance directions. The strict constant-phase replacement gives nearly the same joint-pair result; identification here therefore comes mainly from keeping the separate image/flux/delay information rather than resolving the small transition correction. Astrometry alone retains only very small directions in this example, and its near-null spectrum is numerically delicate. The apparent-mass statistic alone retains no measurable independent direction with the admitted free-source/action-monopole family. The analytic point-mass/shear null in Equation (66) separately establishes an exact first-order loss without relying on an SVD cutoff.

Controls vary mass, worldtube radius and the supplied scalar flux. With zero spherical scalar flux the exterior quadrupole is harmonic, $f_2 \propto r^{-3}$; its transverse projected phase is exactly an ordinary compact quadrupole and the retained response vanishes to numerical precision. Nonzero

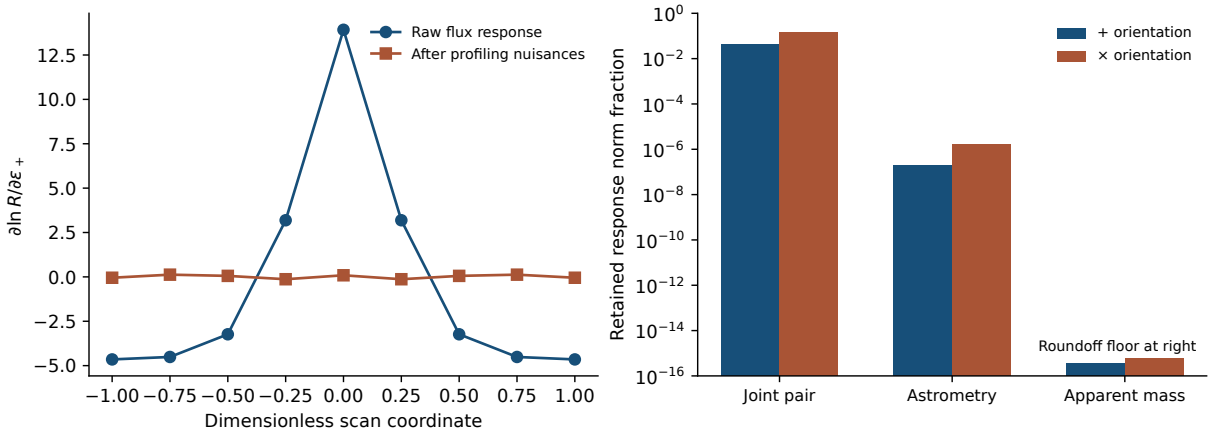


Figure 3: Information retained by the declared broad scan. Left: the flux component of the + response before and after simultaneous nuisance projection in the full data vector. Right: retained column-norm fractions for three data choices; the apparent-mass residual is at roundoff. Comparisons use the stated dimensionless metric and nuisance family, not a telescope covariance or a universal observability theorem.

fluxes 0.1, 1 and 3 times the proxy give a screened response with the same broad-scan information pattern, as do the tested 10^6 and $10^{11} M_\odot$ examples with declared weak-field worldtubes. These are conditional boundary families, not alternative theories or inferred black-hole charges. Allowing an independent center at every configuration reduces the joint information but does not remove both transverse directions in this restricted family. Arbitrary time-dependent quadrupoles would be a different, less constrained problem.

7.3 Perturbative and numerical control

The physical state example is chosen by an explicit small parameter,

$$a_+ = \frac{\eta x_0 |p(x_0)|}{6}, \quad a_\times = 0, \qquad \eta = 10^{-3}, \quad x_0 = r_0/L_0. \tag{70}$$

The factor six is the quadrupole angular eigenvalue; the numerical checks also evaluate actual Hessian ratios and the quadratic residual in the retained nonlinear scalar equation. This is a supplied perturbatively small state, not a BHSM-selected amplitude or an upper bound on astrophysical deformations. It gives $a_+ \simeq 1.38 \times 10^{-16}$ and $\epsilon_+ \simeq -8.89 \times 10^{-13}$.

At source $(0.9, 0.6)$ the retained model gives a largest coordinate shift of about 9.4×10^{-5} nanoarcsec and $\delta \ln R \simeq 2.88 \times 10^{-12}$. The formal matched delay shift is approximately 5.1×10^{-12} s. That last small number is a value in the truncated static thin-lens model, whose leading delay term vanishes; it is *not* a physically certified picosecond forecast. Uncomputed time-dependent and matching corrections require their own comparison before interpreting such a subleading delay.

The code checks the radial equation against the special-function solution, the dipole translation mode, independent original-z versus transformed line integrals, finite-difference projected derivatives, boundary energy normalization, and finite nonlinear primary-image continuations. At test amplitudes decreasing from 10^{-3} to 3×10^{-5}, the central-difference error in the full response decreases quadratically to about 1.6×10^{-7}, with image stationarity residuals below 10^{-14}. Those larger optical test amplitudes test the lens derivative; they do not validate the small-state scalar linearization at an enlarged boundary amplitude. Independent symbolic/Cartesian and polynomial-branch checks verify the estimator operator and its nulls. All are internal AI-assisted checks.

Quintic interpolation uses analytic projected derivatives at logarithmic nodes; measured midpoint discrepancies and quadrature refinements are recorded as numerical diagnostics, not interval certificates. The formal infinite exterior is matched to the frozen local background. At a 1 Mpc cutoff, explicit far-tail bounds give a monopole deflection tail below 9×10^{-11} of the ordinary deflection and a transverse quadrupole phase tail below 8×10^{-11} pc per unit boundary amplitude. These bound omitted tails only. They do not bound every weak-field, cosmological or effective-theory correction.

8 Materials and methods: the runnable Blackhole Tracker product

The released research product makes the compact-lens measurement task operational. A local browser app, command line, importable Python package and four executed notebooks provide separate versioned resolved-pair, native-photometry and joint astrometry/parallax contracts. This is an offline analysis tool; the resolved mode described here takes supplied primary-pair measurements. The native route and its measured-data pilot are treated in Section 10. Its ordinary-lens mode estimates the mass, center and motion of the BHSM-described exterior, with the explicit boundary dictionary of Section 2. Its optional R1 mode evaluates the conditional exterior and the admission bound of Section 4. Established lens software already supports sophisticated lens inference: `lensmodel`, `lenstronomy` and `PyAutoLens` [19–21]; the moving-lens `BAGLE` framework is a particularly close predecessor [22]. No first mass/motion tracker or general replacement for those tools is claimed.

The comparison is specific. Cubic-Galileon multipole calculations [8, 9] supply screening powers and Green functions; general STF lens projection [11] supplies multipole transfer methods. Here those ingredients are composed with the adopted R1 coefficients, a normalized boundary/conormal pairing and a sufficient state cap, followed by explicit delay/statistic nulls. The extended-source lens packages above address broader image reconstruction and lens inference; `BAGLE` models moving microlenses with astrometric, photometric and parallax information. This product instead takes resolved primary pairs, their geometric flux ratio, supplied delays and a full covariance, and reports the conditional BHSM dictionary and R1 admission/information assessment. It supplies no general extended-image reconstruction, survey acquisition or observational validation of those assumptions. No superiority or priority follows from these scope differences.

Computational methods and generative AI. The reproducible software is Blackhole Tracker, version 0.2.0rc4; the versioned source, environments and executable checks are supplied with the release. OpenAI ChatGPT and Codex (GPT-6 model family) assisted source recovery, mathematical derivation, code and test generation, numerical analysis, literature checking and manuscript drafting. Synthetic measurements are generated by the released numerical model, not by language-model sampling. Symbolic identities, finite differences, numerical residuals, source hashes and independent internal agent checks test the retained output. These checks do not replace author review or external peer review; their scope and limitations are reported below.

8.1 Measurement contract and likelihood

An input supplies angular-diameter distances D_l, D_s, D_{ls}, redshifts, a reference observer epoch, and N epoch/source rows, with $3 \le N \le 80$. Each row contains two resolved image positions in mas, $\ln(|\mu_+|/|\mu_-|)$ and the negative-parity minus positive-parity arrival-time difference in seconds. The labels refer to parity, not a brightness ordering; a perturbed flux ratio may be below unity. The covariance is the full positive-definite $6N \times 6N$ matrix in row-major observable order. Neither time sorting nor diagonalization is applied. Correlated astrometric, flux and timing errors therefore

remain coupled under Cholesky whitening. A provenance field explicitly declares synthetic or observed measurements. Validation of that field and the numerical schema does not verify an observation's origin.

With supplied parallax factors $\boldsymbol{P}_i$, the lens center is

$$\boldsymbol{c}_i = \boldsymbol{c}_0 + \boldsymbol{\mu}_{\rm obs}(t_i - t_0) + \frac{1000}{D_l/\mathrm{pc}}\boldsymbol{P}_i \text{ mas}. \tag{71}$$

Omitting these factors means a declared zero observer-parallax term. Distances, the astrometric frame and timing calibration are treated as known; they are not silently inferred from the lens data. The source position has two free coordinates at every row, even when several rows represent repeated observations of one source. Source trajectory truth is never supplied to the fitter. The ordinary model contains a point mass and constant center motion, with selectable shear or the larger family of shear, convergence and compact quadrupole. It solves the nonlinear lens equation for the continued positive/negative primary branches, rejects critical or incorrectly ordered branches, and uses their full magnification determinants and Fermat delay.

The optimizer minimizes

$$\chi^2(\vartheta) = \|L^{-1}[d - f(\vartheta)]\|_2^2, \qquad LL^T = C. \tag{72}$$

An analytic point-lens delay/flux initializer supplies a computational mass scale, followed by deterministic perturbed starts and bounded nonlinear least squares. The default mass search domain is $[10^{-3}, 10^{12}]M_\odot$, exposed in the input rather than confused with a detection prior. A conservative slow-motion domain is declared separately. Final solutions undergo primary-branch, actual-impact weak-field and motion checks; R1 additionally requires its exterior worldtube and state-admission conditions. Local covariance follows the whitened Jacobian SVD, without rescaling the supplied C by reduced χ^2. Null directions have unavailable uncertainties. Center-track bands propagate the center–motion cross covariance. Reoptimized one-coordinate likelihood profiles are available from the command line and notebook; their usual $\Delta\chi^2$ reference values do not guarantee nonregular coverage.

The reviewed implementation optimizes near a data-derived epoch without finite numerical boxes on center or source positions. It reports the center at the supplied epoch using $\boldsymbol{c}_{\rm rep} = \boldsymbol{c}_{\rm opt} + \boldsymbol{\mu}(t_{\rm rep} - t_{\rm opt})$. The full identified-subspace covariance and null directions transform with the reporting Jacobian, including center–motion cross terms. Unidentified physical covariance is unavailable; a truncated inverse is not reported as precise motion uncertainty. Weak but identifiable motion retains finite covariance. Numerical reference mass changes do not alter the physical position domain or the separately fixed physical bounds of compact multipoles. Persisted basis metadata prevent reinterpretation of legacy internal coordinates.

Each fixed-parameter profile genuinely searches admissible nuisance states under the public weak-field, worldtube, motion and conditional-R1 constraints. Interior and boundary optima remain distinct from numerical failure and unresolved physical-domain searches. A failure to find an admissible state does not prove infeasibility. Only admitted solved rows relative to an admitted reference minimum can supply a physical likelihood crossing; unresolved rows have unavailable likelihood values.

8.2 Product decisions and the conditional extension

`FIT_REJECTED` denotes solver failure, unacceptable residuals at the declared $p < 0.001$ guard, or failure of a physical-domain check. `INSUFFICIENT_INFORMATION` denotes unavailable or very weak mass information. Otherwise, `PARTIALLY_IDENTIFIED` reports weak motion or a global parameter at its search bound. `TRACKED` requires a supported mass and a two-dimensional local motion Mahalanobis radius of at least five, together with the other checks. This is a transparent

product decision rule within the supplied model and covariance; it is not a black-hole classification or a universal false-alarm probability. Every result retains its full residuals, assumptions and diagnostics.

In R1 mode the spherical flux is either the explicitly labeled fixed regular-body proxy or a fitted conditional matching datum. The latter is bounded to the declared fitted domain $0.05 \leq \chi \leq 20$; discrete runtime comparisons do not certify that entire interval. Neither option is called horizon matched. After fitting the monopole and ordinary nuisances, unit-amplitude transverse responses are whitened and projected off the fitted Jacobian and the visible axial response. If their singular values are $s_1 \geq s_2$, the diagnostic compares $\mathcal{A}_{\max} s_j$ with unity. A rank loss or best admitted response below unity returns `UNRESOLVED`. The alternative label, `INFORMATION_AVAILABLE`, denotes local conditional sensitivity, not a detected deformation. A formal unconstrained state outside the cap is explicitly marked outside the admitted family. The product does not inflate it into a physical fit.

For the noisy nearby fixture below, the ordinary fit gives $M = 99947.2 \pm 38.7\, M_\odot$ and $\boldsymbol{\mu} = (0.06667, -0.03808)$ mas/yr with local component errors $(0.00119, 0.00123)$ mas/yr; $\chi^2 = 36.39$ for 29 degrees of freedom. These are one synthetic realization's outputs. For R1 analysis with the fixed regular-body flux proxy and shear nuisances, that fixture can retain the ordinary track while returning an unresolved envelope: the admitted state cap is about 5.34×10^{-21} and the largest projected admitted local response-to-noise is about 2.33×10^{-14}. Thus making the tracker usable does not supply evidence for the proposed BHSM deformation signal.

The app supports input inspection, validation, fitting, track and observable plots, uncertainty/null diagnostics and JSON/HTML export. Both app and command line export a complete replay analysis: original input with full covariance and calibrations, fit configuration, installed software/source hashes, fitted parameters, predictions, diagnostics, interpretation, plots and a SHA-256 manifest. The command `python -m bhtracker replay analysis.zip --output replayed` verifies hashes and repeats resolved/native-photometry inference. The resolved/native refit comparison requires unchanged scientific status and prediction differences below 10^{-3} of each supplied marginal standard deviation; this is not a rescaling of measurement error or a statistical detection threshold. For the separately versioned joint route, replay verifies the saved nonlinear state, profiled nuisances and predictions with its pinned scientific source. It does not repeat optimization or certify noise adequacy, mass support or black-hole classification. The first notebook reproduces mass/motion inference, a mass profile and the weak-motion control; the second evaluates conditional response and admission. The third preserves native unresolved photometry. The fourth verifies the separately versioned joint likelihood, source-pack import and archived measured-state replay; its successful execution does not accept the measured benchmark's mass. Raw theory inputs are packaged and hash checked, so these interfaces do not depend on a private archive path or the author's checkout. A separate stellar-mass example adapts rounded parameters from Lu et al. [22] into a declared synthetic primary-pair test with a chosen observer ellipse. Its positions, fluxes and added delays are generated by this product. No `BAGLE` engine was run, and no observed data or real ephemeris was validated. The independent ordinary-lens checks instead evaluate the analytic point-mass equations in physical units separately from the package's lens helpers. The repeated-noise experiment tests recovery and local interval coverage under the product's own declared synthetic model.

Versioned historical calibration. The following resolved-pair calibration retains its original 64-draw experiment and separate 128-draw follow-up, source versions and failed controls. It is historical evidence for that model and implementation, not a repeated calibration of the revised rc3 kernel or native unresolved-photometry likelihood.

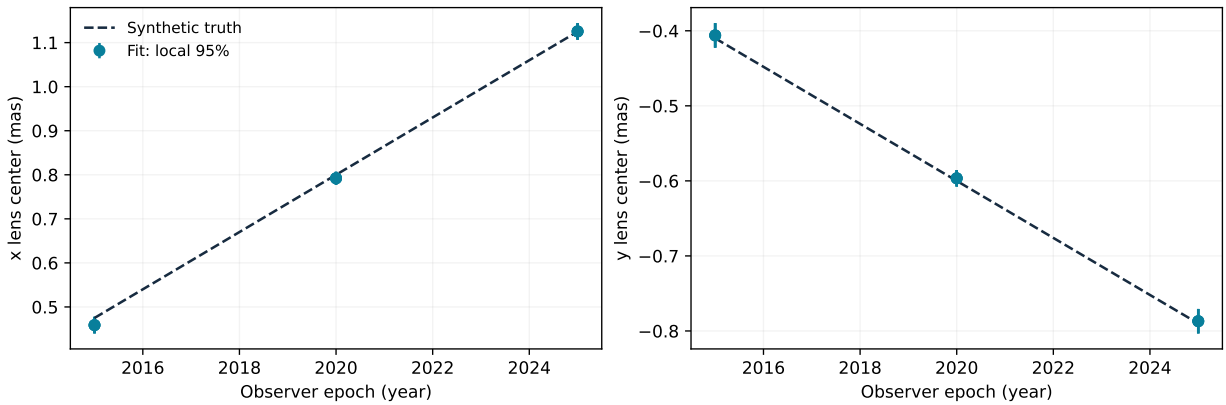


Figure 4: The noisy nearby synthetic fixture: recovered lens-center coordinates with local 95% marginal intervals propagated from the full center–motion covariance, against truth used only for scoring. Repeated rows at each epoch overlap. This is a ten-year three-source product test, distinct from the broad single-track information scan.

9 Results: nonlinear product validation with known covariance

The software product is tested separately from the preceding 54-observable theory experiment. Its ordinary compact-lens inference fits mass, reference center and observer proper motion, two shear components, and two free source coordinates at every row. Distances, redshifts and the absolute astrometric frame are supplied calibrations. The baseline validation is entirely synthetic; its adopted errors do not specify an instrument capability. Optional R1 information retains the separate conditional-exterior and physical-amplitude restrictions. The frozen nearby fixture contains a $10^5 M_\odot$ lens at $D_l = 1$ Mpc, three background source positions sampled at each of three epochs over ten observer years, and true motion $(0.065, -0.038)$ mas/yr. The nine rows supply 54 image-coordinate, log-flux-ratio and delay measurements. Their declared base noise scales are 0.02 mas, 0.005 and 3×10^{-4} s, respectively, with supplied astrometric and timing correlations. The full covariance C is retained in $\chi^2 = (d - f)^T C^{-1} (d - f)$; errors are not rescaled by the fitted reduced χ^2. These inputs test a resolved primary-pair model with free source positions, rather than assume a known unlensed source track.

The noiseless recovery control (C6) uses the data-derived initializer and two fixed perturbations, including mass factors 1.5 and 0.65 and shifts of center, motion, sources and shear. Each fit uses three internal starts. All recover the mass, center and motion within the frozen relative or reference-scale tolerances 10^{-5}; the largest χ^2 is 3.01×10^{-23}, below the fixed 10^{-6} criterion. Truth is used to score recovery, not supplied as an initializer. This checks nonlinear solver recovery for this fixture, not global likelihood uniqueness.

9.1 A fixed repeated-noise calibration

Control C7 draws 64 independent Gaussian noise vectors using the full C, a single frozen PCG64 stream seeded 4932026, and one start per ordinary-lens fit with shear admitted. Every realization remains in the record and coverage denominator; an exception, rejected fit or unavailable interval would count as a miss. All 64 fits converge and return the product's `TRACKED` status, with no excluded failures. The reported intervals are local known-covariance Gaussian intervals, not reoptimized likelihood profiles. For mass we use $\widehat{M} \exp[\pm 1.959964\, \sigma_M / \widehat{M}]$; center and motion use $\widehat{\theta} \pm 1.959964\, \sigma_\theta$.

None fails the preregistered gross-miscalibration guard (Table 4). The exact 95% binomial interval for 61/64 coverage is $[0.8691, 0.9902]$, and for 59/64 it is $[0.8270, 0.9741]$: 64 draws provide a limited calibration check. The receipt also retains parameter bias and scatter. In particular, the

Table 4: Frozen 64-draw local 95% interval coverage. Each exact two-sided binomial test has null coverage 0.95 and rejection level $\alpha = 0.01$. All five tests are reported without a multiple-testing adjustment.

Coordinate	Covered / 64	Fraction	Exact p
Mass M	61	0.9531	1.0000
Center x	59	0.9219	0.2532
Center y	61	0.9531	1.0000
Motion x	61	0.9531	1.0000
Motion y	61	0.9531	1.0000

mean y-motion error is 3.98×10^{-4} mas/yr, about 2.75 estimated standard errors of its ensemble mean. Passing the frozen coverage test does not establish negligible bias or universal coverage.

9.2 A bounded investigation of the mean offset

Before generating follow-up data, a separate protocol fixed an independent analytic linear-noise control, tighter checks of original indices 0–7, and, conditional on those checks, exactly 128 further draws from PCG64 seed 202610210128. The original receipt, thresholds and 64-draw table are unchanged; the two ensembles are not pooled. The protocol's initially announced version was clarified before execution to specify that mean diagnostics use all finite returned estimates, including those returned with rejected decisions; such decisions remain interval-coverage misses. The revision history and both hashes are retained in the risk receipt.

The independent 25-column point-lens, shear and free-source Jacobian uses stationary-image derivatives in physical units. Its whitened relative discrepancy from the production Jacobian is 2.50×10^{-11}, and the largest discrepancy of the five covariance diagonals is 2.91×10^{-11}, below fixed bounds 10^{-5} and 10^{-4}. Its original-sample mean y-motion error is 3.98022×10^{-4} mas/yr, versus the nonlinear 3.97992×10^{-4} mas/yr: their difference is only -0.000199 analytic standard errors of the ensemble mean. The offset is therefore already present in a first-order response with exactly zero expected error. The largest mean nonlinear-minus-linear discrepancy among all five coordinates is 0.00509 analytic standard errors, below the fixed 0.1 concern level. Eight tighter refinements change estimates by at most 4.57×10^{-8} local standard deviations and improve χ^2 by at most 1.55×10^{-12}, below the respective fixed bounds 0.01 and 10^{-6}. These checks find no material covariance or local convergence discrepancy in this fixture.

All 128 optimizers converge and return finite estimates; 126 decisions are `TRACKED` and two are `FIT_REJECTED` at the unchanged goodness-of-fit guard $p < 0.001$. All remain in the record and denominator (Table 5). Neither the coverage nor local mean-error guards reject. The follow-up y-motion mean is 9.13×10^{-5} mas/yr, or 0.879 empirical ensemble standard errors. The mean diagnostics include finite estimates from the two rejected decisions as specified; they are not admitted tracks. A labeled post-hoc projection of the already-used noise through the independent

Table 5: Separate fixed 128-draw follow-up. Rejected decisions count as misses. Coverage tests retain null 0.95 and per-coordinate $\alpha = 0.01$. Mean z uses the known truth linear covariance; the prespecified five-coordinate mean guard has Bonferroni familywise $\alpha = 0.01$.

Coordinate	Covered / 128	Exact coverage p	Mean z
Mass M	118	0.1513	−0.4555
Center x	120	0.5383	−0.2571
Center y	119	0.3045	−0.1777
Motion x	120	0.5383	−0.7367
Motion y	121	0.6872	0.8394

Table 6: Controls with scientifically meaningful weak or null outcomes.

Control	Quantity	Outcome
C8: short cosmological track	Motion SNR 0.00482	Unresolved motion
C11: shear omitted	$\chi^2 = 28297.8$	Fit rejected
C11: shear admitted	$\chi^2 = 1.95 \times 10^{-22}$	Baseline recovered
C9: harmonic / compact pair	Four columns, rank two	Two exact nulls

linear model rejects exactly the same two draws, with $\chi^2 = 65.4492, 70.3639$ on 29 degrees of freedom, compared with nonlinear $65.4439, 70.3620$. Two such rejections are uncommon under an ideal size-0.001 guard: $\Pr[\text{Binomial}(128, 0.001) \geq 2] = 0.00748$. This limitation is retained without changing thresholds or replacing data. The study does not calibrate an operational false-rejection rate.

The bounded execution uses 145 solves and leaves the reserved correction stream unused. No production bias correction or recentering is applied. These results are consistent with finite-sample fluctuation of the highlighted offset in this fixture, while preserving its observed value. They support limited synthetic local-interval calibration, not proven unbiasedness, universal coverage or observational sensitivity.

9.3 Weak information, model mismatch and finite cause ambiguity

The negative controls preserve unsuccessful scientific outcomes (Table 6). The ten-year cosmological fixture (C8) has true motion $(5, -3) \times 10^{-5}$ mas/yr and returns `PARTIALLY_IDENTIFIED`, with motion signal-to-noise 0.00482 and component errors $(0.0121, 0.0124)$ mas/yr. Its mass information does not license a resolved motion claim. For a separate conventional-shear fixture (C11), omitting shear gives a rejected fit at the frozen goodness-of-fit criterion $p < 0.001$; admitting shear recovers the declared $(0.03, -0.018)$ coefficients. Residual structure is therefore not assigned automatically to an R1 envelope or a new physical cause.

For C9 the zero-flux harmonic boundary response is exactly an ordinary compact r^{-2} quadrupole, with coefficient $-\widehat{\beta} N_2 r_0^3 / [3 m_{\text{ref}} (r_E^{\text{ref}})^2]$ per boundary amplitude, in the declared reference lens units. Three finite operator comparisons and compensating coefficient changes agree in potential, gradient and Hessian to 2.22×10^{-16}. Four combined compact/boundary columns have rank two. Large formal coefficients are confined to this operator identity test; public prediction separately rejects that nonadmitted state. They are not enlarged physical signals. The current nonlinear fit does not expose a joint free fit of both degenerate amplitude pairs, so separate-amplitude interval calibration is not claimed. These controls accompany the released independent interface, physical-unit, covariance and admission tests. Versioned JSON receipts retain every draw, initializer, criterion, runtime and SHA-256 source hash; the R1 provider inputs are pinned and checked on loading. The scripts `calibrate_product.py` and `check_finite_harmonic_degeneracy.py` reproduce the controls. The validated scope is the declared synthetic compact-lens families and local intervals. Unknown calibration, broader environmental freedom, critical image configurations, observational sensitivity and horizon identification remain outside this validation.

10 Native unresolved observations: a frozen 100-event pilot

10.1 A separate observation contract and its identifiable quantities

The resolved-pair experiments above are not the observation model for an unresolved survey light curve. Release candidate 0.2.0rc3 adds a separate, versioned native-photometry route. An input row retains its original time, apparent magnitude, uncertainty, instrument/filter, source

file hash and source-line identity. Catalog coordinates, aliases, calibration notes and documented warnings remain provenance. No image positions, primary flux ratios, interimage delays, distances or covariance are manufactured from those rows.

For a rectilinear point-source point-lens (PSPL) event [23], the dimensionless source–lens separation and unresolved magnification are

$$u(t)^2 = u_0^2 + \left(\frac{t - t_0}{t_E}\right)^2, \qquad A(u) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}. \tag{73}$$

For instrument j, $F_j = F_{s,j}A + F_{b,j}$. Writing the baseline magnitude as $m_{b,j}$ and the baseline source fraction as $f_j = F_{s,j}/(F_{s,j} + F_{b,j})$ gives the native-magnitude prediction

$$m_j(t) = m_{b,j} - 2.5\log_{10}[1 + f_j(A(u(t)) - 1)], \qquad \chi^2 = \sum_i \frac{[m_i - m_{j(i)}(t_i)]^2}{\sigma_i^2}. \tag{74}$$

This likelihood assumes independent Gaussian errors in the supplied native magnitudes. It preserves the supplied errors; it does not infer an unprovided cross-row covariance or establish that the independence assumption is correct. Per-instrument baselines absorb independent photometric zero points without treating published catalog parameters as new measurements. The catalog solution initializes one of twelve deterministic starts and contributes no likelihood term.

The optical model has the same BHSM Einstein/GHY interpretation as Section 2: one effective exterior acts once. Unresolved PSPL photometry, however, generally constrains only event time, dimensionless impact parameter, timescale and blending combinations, to the extent admitted by its sampling and degeneracies. In the usual angular convention,

$$\theta_E^2 = \kappa M \pi_{\rm rel}, \quad \pi_E = \pi_{\rm rel}/\theta_E, \quad M = \frac{\theta_E}{\kappa \pi_E}, \quad \mu_{\rm rel} = \theta_E/t_E, \qquad \kappa = \frac{4G}{c^2\,\mathrm{AU}}. \tag{75}$$

The baseline fit measures neither θ_E nor π_E [24]. A long t_E can arise from mass, relative distance or slow relative motion. Mass, lens distance, absolute lens motion, Schwarzschild radius, and the mass-derived boundary dictionary therefore remain unavailable for every pilot event. No core state or deformation is inferred from ordinary PSPL agreement. The optional astrometric centroid and geocentric-parallax primitives have analytic convention and input-admission tests [25, 26]; this pilot is not an end-to-end joint astrometric mass analysis.

10.2 Data integrity, selection and protocol freeze

The supplied observations are the optimized OGLE-III Class A catalog and its native light curves [27], together with the separate 2022 OB110462 benchmark deposit [28]. All 109 supplied raw-file hashes verify. The pilot exactly reproduces the frozen 100-event selection: all 75 catalog events with published $t_E \geq 150$ d, and 25 hash-selected controls from 2272 eligible events with $10 \leq t_E < 50$ d, in a 3560-event parent catalog. These are historical published events. Catalog selection and long-duration enrichment prevent a completeness, black-hole abundance or population-rate inference. Although the original standard-event selection disfavors conspicuous parallax/exotic events, later alias/coordinate checks find four exact members of the published 59-event parallax sample [29]; it is incorrect to describe the cohorts as disjoint.

All 76 344 numeric pilot rows enter the baseline and both sensitivity fits. The loader records four documented source annotations, retains OGLE3-ULENS-0337, and applies no clipping, thinning, averaging, duplicate deletion, event replacement or second application of published error corrections. Native HJD relative-day arithmetic is sufficient for this rectilinear screen; its underlying clock scale is retained as unspecified, rather than guessed for a precision ephemeris calculation. The downloaded bytes are unchanged after analysis.

Protocol `BHSM-OGLE3-native-pilot-20261022-v1` was frozen at 19:50:27 UTC on 22 October 2026 before measured-light-curve fits. Its SHA256 is `b1e75ea708343bad1207ac180229058f093164e877cea68ac13bedfc3feed498`; the ordered event-ID digest is `0cf5dbf714a99567ccfc8448f53560333ed4597115a2801138ac707926a8a4a7`. This is an internal frozen analysis, not external preregistration. The protocol pins the raw data, driver, likelihood and source modules; the completed results and later literature annotations have separate manifests. The original kernel is retained for replay. Post-pilot astrometry docstring clarifications change no executable syntax, and the native-photometry and observation modules remain byte-identical. The final rc3 theory-source update to revision `705d4c7e` is a separately recorded provenance change; the pilot retains its original interpretation pins and unchanged PSPL objective.

10.3 Numerical admission, uncertainty and limited validation

The fit uses bounded trust-region least squares with an analytic Jacobian, twelve starts, 1500 evaluations per start, a fixed seed, linear loss and declared stopping tolerances. Search bounds are $0.1 \leq t_E \leq 10^4$ d, $10^{-4} \leq u_0 \leq 10$, $10^{-6} \leq f_j \leq 1$, t_0 within the data range extended by one observed span on each side, and each baseline within ten magnitudes of its instrument median. These are explicit computational/model bounds, not measured physical limits. An active bound flags nonregular local uncertainty. A column-normalized Jacobian SVD uses relative cutoff 10^{-7}; unidentified directions yield unavailable public uncertainties rather than zero-error estimates.

Every event also has a nuisance-reoptimized timescale profile. Seventeen log-spaced values cover $[t_E/4, 4t_E]$, clipped to the declared search range, with the fitted point included. Four deterministic source-fraction seeds reoptimize all remaining nuisance coordinates at each trial. Adjacent admitted converged rows bracket a $\Delta\chi^2 = 1$ reference crossing, refined by up to sixteen logarithmic bisections with new nuisance fits. A failed row blocks that crossing; a finite grid endpoint is not a physical confidence limit. Both crossings are found for 91 events; nine remain grid-interval unresolved. These are likelihood reference levels, not calibrated coverage intervals, especially for an inadequate model.

Two predeclared sensitivity fits are retained for all 100 events: uniform native-error multiplication by 1.25, and a signed-blend phenomenology with source fraction up to ten. The latter represents possible calibration/background offsets; it is not negative physical stellar luminosity. Neither replaces the baseline. Diagnostics retain multistart alternatives, active bounds, local rank, constant/linear-trend BIC comparisons, runs and lag-one residual statistics, seasonal residual offsets, and instrument-specific residuals.

Before real-data inference, synthetic injections on two actual sampling patterns test the new likelihood. Noiseless $t_E = 200$ d and 25 d injections recover their parameters with $\chi^2 < 2 \times 10^{-23}$; eight independent-noise draws at each cadence test recovery and error scaling. A weak-source injection exposes the blending–timescale profile degeneracy. Sixteen noise trials do not establish interval coverage, false-discovery control or black-hole recovery. They are separate from the historical 64- and 128-draw resolved-pair calibrations above; those receipts retain their original code versions and are not new-kernel coverage tests. Focused F1–F3 correction verification comprises 75 passing tests of physical reference invariance, admitted nuisance profiles and unidentified-motion reporting; this count is not a new observational calibration.

10.4 All-event outcomes and bounded follow-up ranking

The run completes all 100 events without changing the cohort. Eighty-nine are `PHOTOMETRIC_MODEL_INADEQUATE`; eleven receive `PHOTOMETRIC_FIT` because the specified independent-error χ^2 screen does not reject them. The latter label is not full scientific adequacy: residual structure, boundary effects and alternative models remain separate tests. The long-duration stratum contributes 67 inadequate and eight non-rejected fits; the controls contribute 22 and three,

Table 7: The only two events passing the frozen archival-follow-up screen. Profile limits are nuisance-reoptimized $\Delta\chi^2 = 1$ reference crossings in days, without a coverage claim. Mass is unavailable for both.

OGLE3-ULENS	Rows	χ^2/ν	Fitted t_E (d)	Profile (d)
0585	734	748.247/729	268.17	[202.91, 344.96]
3092	739	805.423/734	187.73	[170.06, 208.64]

respectively. Failed or inadequate outcomes remain in the released all-event table and in every denominator.

The frozen archival-follow-up screen additionally requires a finite locally identified $\sigma_{t_E}/t_E < 0.3$, both profile crossings, confirmed multistart results, no active bounds, no prescribed runs or seasonal residual flag, $\Delta\mathrm{BIC} > 10$ relative to a linear trend, and timescale stability in both sensitivities. Ranking then sorts lexicographically by $|\log \chi^2_\nu|$, profile fractional half-width, decreasing timescale and event ID. These operational criteria select two historical objects (Table 7); they are not black-hole probabilities or an externally validated discovery statistic. Figure 5 shows both selected light curves and their residuals, retaining every native observation.

The fitted baseline source fractions, 0.180 and 0.195, emphasize that blending remains consequential. OGLE3-ULENS-0585 is catalog source `BLG122.6.83113`; the supplied catalog has no EWS alias. OGLE3-ULENS-3092 is `BLG225.6.198210`, independently associated with OGLE-2007-BLG-057 by the official EWS coordinates (separation 0.75 arcsec). It is also MOA-2007-BLG-027, discussed by Yock [30]; its prior high-magnification event and incomplete peak coverage are not a new discovery. The historical EWS and optimized catalog star identifiers belong to different reductions and are both retained. Bounded alias/name/coordinate searches locate no specific earlier lens-interpretation paper for 0585, but absence from this search does not establish novelty. Neither object has measured mass here; both are candidates for archival reanalysis and imaging feasibility, not current alerts or credible new black-hole classifications.

The four exact prior-parallax matches in the full pilot are 0264/PAR-07, 0568/PAR-21, 2122/PAR-37 and 3404/PAR-17. In particular, 0264 was already proposed as a dark-remnant candidate in the 2016 study. The later Gaia-informed analysis changes its conditional inference [31]; a prior-dependent literature mass is not imported as a mass measurement by this PSPL fit. Machine-readable literature annotations preserve these associations alongside the unchanged frozen results. The run yields zero mass-supported black-hole candidates and zero new black-hole interpretations.

10.5 The historical known-system benchmark and its stopping point

OB110462 (OGLE-2011-BLG-0462/MOA-2011-BLG-191) is analyzed only with the corresponding Zenodo 6607578 version 1.0.0 release [28, 32, 33], separately from the pilot. The deposit's requested OGLE attribution is also retained [34]. Its 15 545 OGLE and 46 040 MOA photometric rows are retained. Later analyses use additional observations and revised reductions [35, 36]; their masses and newer measurements are not substituted for the 2022 input version. All five specified PSPL fits converge with confirmed multistart solutions, yet all fail the declared photometric adequacy screen and show time-correlated/seasonal residuals (Table 8). Convergence therefore cannot be presented as recovery of the published black-hole mass. Parallax, source/lens complexity and noise calibration require their own validated likelihoods.

Sixteen HST rows at eight shared visits are ingested with native column labels, but are not used for precision joint inference. Primary-source recovery establishes that the 2022 reduction applies a cross-filter alignment, while a historical BAGLE reader converts private decimal-year UTC epochs to MJD. Neither supplies the missing exact export mapping for these deposited ASCII times. A

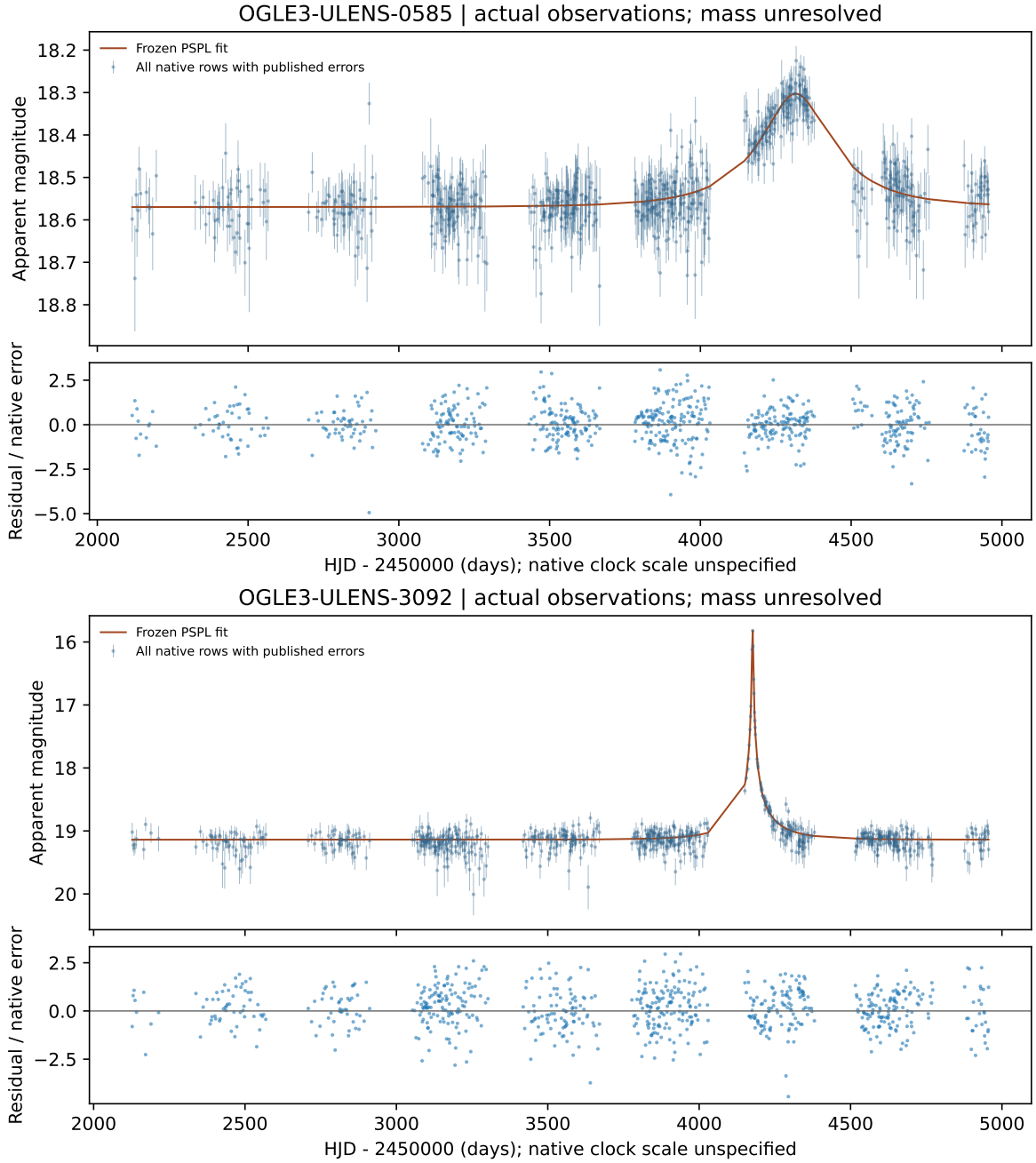


Figure 5: Actual observations of the only two events passing the frozen archival-follow-up screen. Curves are the saved baseline fits; bars are the original magnitude errors, and lower panels retain all standardized residuals. The fits identify conditional photometric combinations, not lens mass or type. These historical events are not new discoveries.

Table 8: Historical 2022-data benchmark: all entries are inadequate photometry-only fits. The final column evaluates the native-error χ^2, including for the alternative weighting sensitivity.

Fit/sensitivity	Fitted t_E (d)	Native χ^2_ν
Joint OGLE/MOA	282.80	2.4515
OGLE only	270.79	3.6754
MOA only	299.06	2.0259
Equal-instrument objective	276.47	2.4532
Signed-blend sensitivity	282.80	2.4515

concrete check against MAST observation metadata finds all eight native numbers outside the matching F606W exposure windows if naively interpreted as MJD UTC. No guessed epoch shift or second filter offset is applied. Astrometric frame/coordinate and covariance conventions also require verification before composition with an ephemeris. The precision joint route consequently reports missing verified metadata and leaves θ_E, π_E and mass unresolved.

Lam et al.'s published default weighting sums the photometric and astrometric log likelihoods; their equal weighting divides each data-group log likelihood by that group's count [33]. The photometry-only equal-instrument sensitivity above omits HST astrometry and does not reproduce that published joint likelihood. This is a version-specific benchmark of data admission and model failure, not a replacement for either published mass analysis. The source/metadata research, archived source hashes, raw-data provenance, all-event table, fit diagnostics, residual plots and replay instructions are distributed with the review material. Raw OGLE observations remain external pending clear redistribution permission; the Zenodo deposit declares CC BY 4.0.

11 A separately versioned joint angular–photometric benchmark

The unresolved pilot does not identify a lens mass. We therefore evaluate a separate angular–photometric route on the previously studied system OGLE-2011-BLG-0462/MOA-2011-BLG-191 (OB110462). This is a known-system benchmark, not an additional member of the frozen 100-event pilot and not a new discovery. Its observations, calibrations, and analysis version are distinct from the historical 2022 release discussed above. Published mass estimates are comparison results; none is supplied as an observation, prior, initial mass, or acceptance criterion. A successful analysis of this one positive system would establish capability on this data version, rather than a population completeness, false-positive rate, or general guarantee of black-hole identification.

11.1 Native data, calibration, and time provenance

The new version preserves all 15,533 author-reduced OGLE rows, with native HJD, magnitude, reported uncertainty, seeing, and original line identity. The primary published time window, HJD 2455376–2457700, selects 12,500 OGLE observations; the excluded rows remain available for the mandatory all-row sensitivity. The 2023 author Table 2 supplies 21 HST filter observations at 11 calendar visits, each with an east coordinate, north coordinate, magnitude, and respective reported errors [35]. There are consequently 12,521 primary photometric observations and 42 angular coordinate values. East denotes the published tangent-plane $\alpha_* = \alpha \cos\delta$ coordinate; a second $\cos\delta$ factor is not applied. These are unresolved observations, without manufactured image positions, flux ratios, delays, distances, or covariances.

The author tables already include the stated neighbor-bias and additive error calibrations. We do not repeat those corrections or reapply the older empirical filter shift. Native errors remain immutable in the input, export, and replay. The OGLE error calibration was designed to represent empirical observed scatter by brightness and detector [37]; it does not provide a unique decomposition into independent and correlated noise. No full observational covariance is supplied. Every fitted correlation or extra calibration term below is therefore an explicit analysis hypothesis.

The HST table provides dates rather than precise combined-measurement epochs. Program, date, filter, exposure count, and duration match 82 archived FLC primary headers. We derive their UTC exposure midpoints, convert them to TDB, and average the model over all corresponding midpoints for each angular datum. These derived times are not represented as the authors' centroid weights. Duration weights and the full exposure-window endpoints are mandatory sensitivities; the largest visit window spans about 6061 s. The OGLE deposit does not specify the underlying HJD clock scale. The primary import explicitly assumes HJD on the UTC clock, inverts the heliocentric light-time correction, and converts to observer JD on the TDB scale. A ±120 s clock

sensitivity is declared; no resolution of the clock scale is implied. No unresolved 2022 ASCII time origin is repaired using the 2023 calendar dates.

The portable source pack contains the exact author table bytes, corrected OGLE file, exposure-header prefixes, mapping, source URLs, licenses, and SHA-256 inventory. Its standard-library helper verifies the pack and can retrieve the minimal header prefixes using checked HTTP range responses; full FLC images are unnecessary. The two author data releases used here declare CC BY 4.0; the HST header material has separate public-data provenance. Those terms do not imply permission to redistribute every other OGLE product. The scientific source, canonical numeric input hash, protocol, and transport files have independent version pins. Native byte hashes survive browser transport unchanged. We credit the OGLE team and Lam and Lu for the author reductions, and the NASA/ESA Hubble Space Telescope and MAST for public WFC3 exposure metadata; no new HST image reduction is claimed.

11.2 Joint equations and identified quantities

We use north–east vectors and define $\boldsymbol{q}$ from the lens to the unlensed source, in units of the angular Einstein radius. With $\boldsymbol{n} = (\cos a, \sin a)$, $\boldsymbol{b} = (-\sin a, \cos a)$, and the heliocentric Earth position projected on the tangent plane in AU,

$$\boldsymbol{q}(t) = u_0 \boldsymbol{b} + \frac{t - t_0}{t_E} \boldsymbol{n} + \pi_E \boldsymbol{p}_\oplus(t), \qquad A(u) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}, \quad u = |\boldsymbol{q}|. \tag{76}$$

Here t_0, u_0 describe the rectilinear heliocentric trajectory, rather than the apparent peak after annual parallax. The direction a remains a fitted parameter at $\pi_E = 0$. For instrument j, baseline magnitude $m_{b,j}$ and source fraction g_j give

$$m_j(t) = m_{b,j} - 2.5 \log_{10}\{1 + g_j[A(u(t)) - 1]\}. \tag{77}$$

For a dark lens and an unblended angular centroid, the image-summed displacement from the unlensed source is [25]

$$\boldsymbol{\delta}(t) = \theta_E \frac{\boldsymbol{q}(t)}{u(t)^2 + 2}. \tag{78}$$

The measured centroid model adds a free source position, relative-frame proper motion, relative-frame apparent parallax $-\varpi_{S,\mathrm{rel}}[\boldsymbol{p}_\oplus(t) - \boldsymbol{p}_\oplus(t_{\mathrm{ref}})]$, and the declared constant filter offsets. The linear angular nuisance parameters, including θ_E, are profiled within their physical analysis domain at each nonlinear state. They are reprofiled when mass is fixed. Cross-filter angular correlation and extra angular jitter are sensitivity assumptions, rather than covariance claimed to be measured.

Independent annual-parallax and angular-scale information can then identify

$$\pi_{\mathrm{rel}} = \theta_E \pi_E, \qquad M = \frac{\theta_E}{\kappa \pi_E}, \qquad \kappa = \frac{4G}{c^2\,\mathrm{AU}} \simeq 8.14385\,\mathrm{mas}\, M_\odot^{-1}. \tag{79}$$

At zero or unidentified π_E, a finite mass is unavailable. Source and lens absolute parallaxes, distance, and absolute proper motion remain unavailable without an independently calibrated reference frame. Neither a long duration nor an angular fit alone supplies the missing mass information.

Within BHSM, this is one compact-lens optical potential counted once. The ordinary lensing fit is interpreted as the observable exterior lens response of the BHSM object; it is not an additional core potential. These data do not identify a horizon, membrane radius, nongeometric core state, or interior equation of state.

11.3 Preserved failures and the prospective noise extension

The first joint version is retained, including its failed adequacy checks. Its native-diagonal fit had photometric $\chi^2 = 12573.2$ for approximately 12,510 degrees of freedom, but residual lag-one correlation 0.130. A satisfactory global quadratic alone therefore did not establish independent noise. Adding an OU covariance to the full supplied diagonal reduced lag-one correlation to 0.0181, but yielded innovation $\chi^2 = 10720.2$ for approximately 12,508 degrees of freedom. This severe lower-tail deficit was absent from 20 generated and refitted controls (quadratic range 12255.9–12747.4). The fitted covariance family was inadequate; an upper-tail-only check would have concealed the failure. The discrepancy is consistent with overcounting marginal scatter, but does not uniquely establish its instrumental cause. The finite-difference coordinate study also exposed a numerical time-step error. Earlier point estimates are preserved as conditional diagnostic outputs, not accepted mass measurements.

These findings motivate version 2, internally frozen after version 1 exploration and before its own measured fits at commit `5bb9419`. This is a prospective post-diagnostic extension, not an external preregistration. The complete candidate set comprises a centered-time diagonal check (G0), a primary native-marginal OU hypothesis (G1), and an explicit fitted calibration competitor (G2):

$$R_{ik} = (1-f)\delta_{ik} + f\exp[-|t_i - t_k|/\tau], \tag{80}$$

$$C_{\mathrm{G1}} = SRS, \qquad S_{ii} = \sigma_i, \tag{81}$$

$$C_{\mathrm{G2}} = S_{\mathrm{eff}} R S_{\mathrm{eff}}, \qquad (S_{\mathrm{eff}})^2_{ii} = k^2\sigma_i^2 + s^2. \tag{82}$$

The fixed domains are $0 \le f \le 0.95$, $0.05 \le \tau \le 100$ d, $0.5 \le k \le 2$, and $0 \le s \le 0.02$ mag. G1 preserves each reported marginal variance exactly. G2 estimates a separate calibration hypothesis; it never relabels its effective errors as supplied observations. These matrices replace only the OGLE covariance block; HST photometry retains its reported diagonal errors. Both minimize the full Gaussian objective

$$D = \boldsymbol{r}^T C^{-1}\boldsymbol{r} + \log\frac{\det C}{\det D_{\mathrm{native}}} + \chi^2_{\mathrm{angular}}, \tag{83}$$

with the exact, potentially negative relative log determinant. A bounded constant offset permits a real least-squares representation and is subtracted from every reported deviance. The OU calculation is checked against dense covariance algebra, including duplicate epochs and the $f = 0$ and calibration-null cases. Changing the numerical time center preserves the physical model; future targets must supply their own physical time bounds and initial guesses.

11.4 Prediction, controls, profiles, and stopping

The fixed diagnostics screen both quadratic tails, lag correlations, and Ljung–Box structure. Native-error, seeing, noise-free model-brightness quartiles, fixed event/off-event windows, and Julian seasons test residual structure with declared multiplicity qualification. Observed-magnitude bins do not gate a naive quadratic check, because they select residual tails. These postfit screens are approximate; generated and refitted controls assess their behavior.

Independent driver review found that the originally specified $\lfloor \mathrm{JD_{TDB}} - 0.5 \rfloor$ day boundary split Chilean observing nights: 154 adjacent train/validation pairs were separated by less than 0.1 d. That original conditional score is retained with its limitation. Before any prediction outcomes, a separately recorded version 2.1 correction fixed the observing-night unit at local mean solar noon, $N = \lfloor \mathrm{JD_{TDB}} + \lambda/360^\circ \rfloor$, with the Las Campanas longitude and $N \bmod 5 = 0$ held out. The corrected set contains 2464 OGLE rows; all HST rows remain in training. Its minimum adjacent cross-partition gap is 0.6063 d, with no pairs below 0.25 d. The official observatory longitude and the driver's approximate site longitude produce identical row identities.

At the same training-only fitted physical and covariance state, the exact conditional photometric quadratic is $Q_{\mathrm{all}} - Q_{\mathrm{train}}$ and the relative determinant is $\log\det C_{\mathrm{all}} - \log\det C_{\mathrm{train}} - \log\det D_{\mathrm{held}}$. This retains GP correlations across the partition. Angular objective terms are excluded from the difference. Independent dense Schur-complement tests agree to 1.2×10^{-14}. Plug-in prediction probabilities do not account exactly for estimated mean and covariance uncertainty and are not Bayesian evidence.

The protocol fixes at least 20 full-cadence generated/refitted controls per OU candidate, their seeds, multistart modes, mass profiles, and mandatory timing, frame, calibration, blend-location, ephemeris, and all-row sensitivities. Twenty controls resolve Monte Carlo tails only to 1/21 and do not certify interval coverage. Mass support requires resolved competing minima and finite profile crossings, adequate residual and prediction checks, and stability across the declared sensitivities. Numerical caps and unresolved searches interrupt support; physical nuisance boundaries preclude automatic Wilks coverage. If every fixed candidate fails, this campaign stops with a conditional or inadequate mass result. Further outcome-driven model revision is not an acceptance requirement.

11.5 Verified campaign outcomes and scope

All 16 starts per candidate converged after the declared continuation, but all three candidates failed baseline adequacy. Table 9 retains their conditional point estimates; these are not accepted measurements and their spread is not an uncertainty interval.

Table 9: Frozen joint candidates after declared continuation. Each angular quadratic is approximately 32 for 29 conservatively counted degrees of freedom. All candidates fail the residual-structure gate.

Candidate	Conditional $M/M_\odot$	Photometric Q/ν	Lag one	Failed bins
G0	6.546	12568.56/12510	0.1295	15
G1	7.103	12598.17/12508	0.0198	15
G2	18.061	12528.74/12506	0.0464	8

G0 retains strong serial dependence. G1 removes most short-lag structure and passes the approximate Ljung–Box screen, but fails native-error, model-brightness, seasonal, and event/off-event residual checks. G2 fails brightness, seeing, and seasonal checks, has Ljung–Box probability 6.77×10^{-5}, and reaches the artificial 100 d OU timescale cap. Its lower objective does not establish an adequate likelihood or a physical timescale limit. Every global quadratic is apparently acceptable; the failed structure demonstrates why those scalar values cannot establish trustworthy physical uncertainty.

The corrected whole-night conditional quadratics are 2647.64 (G1) and 2540.41 (G2) for 2464 held-out rows, with approximate two-sided probabilities 0.01038 and 0.2771. Both clear the declared 0.01 plug-in screen, G1 narrowly. Their relative Gaussian predictive deviances are 2523.02 and 2245.68. Prediction success cannot override the failed full-data and training residual structures or the G2 numerical cap. Independent reevaluation of the three saved baseline states and two saved prediction scores reproduces every checked quadratic, determinant, and score exactly without refitting. The original midnight-boundary diagnostics are also retained: for 2467 held-out rows, G1 has $Q = 2710.21$ and approximate probability 0.000768, while G2 has $Q = 2596.51$ and probability 0.06846. These values do not supply a whole-night validation, because that partition splits actual observing nights.

Both fixed 20-control sets are complete. None of the 40 generated/refitted cases fails a declared residual-variance bin; 19 of 20 in each family pass the overall screen, with the remaining case rejected by its angular quadratic. Each generated control family's pass fraction 19/20 has an exact 95% binomial interval $[0.751, 0.999]$ for its local generating-model pass probability, rather than mass-interval coverage. Each measured lag-one correlation exceeds all 20 corresponding generated

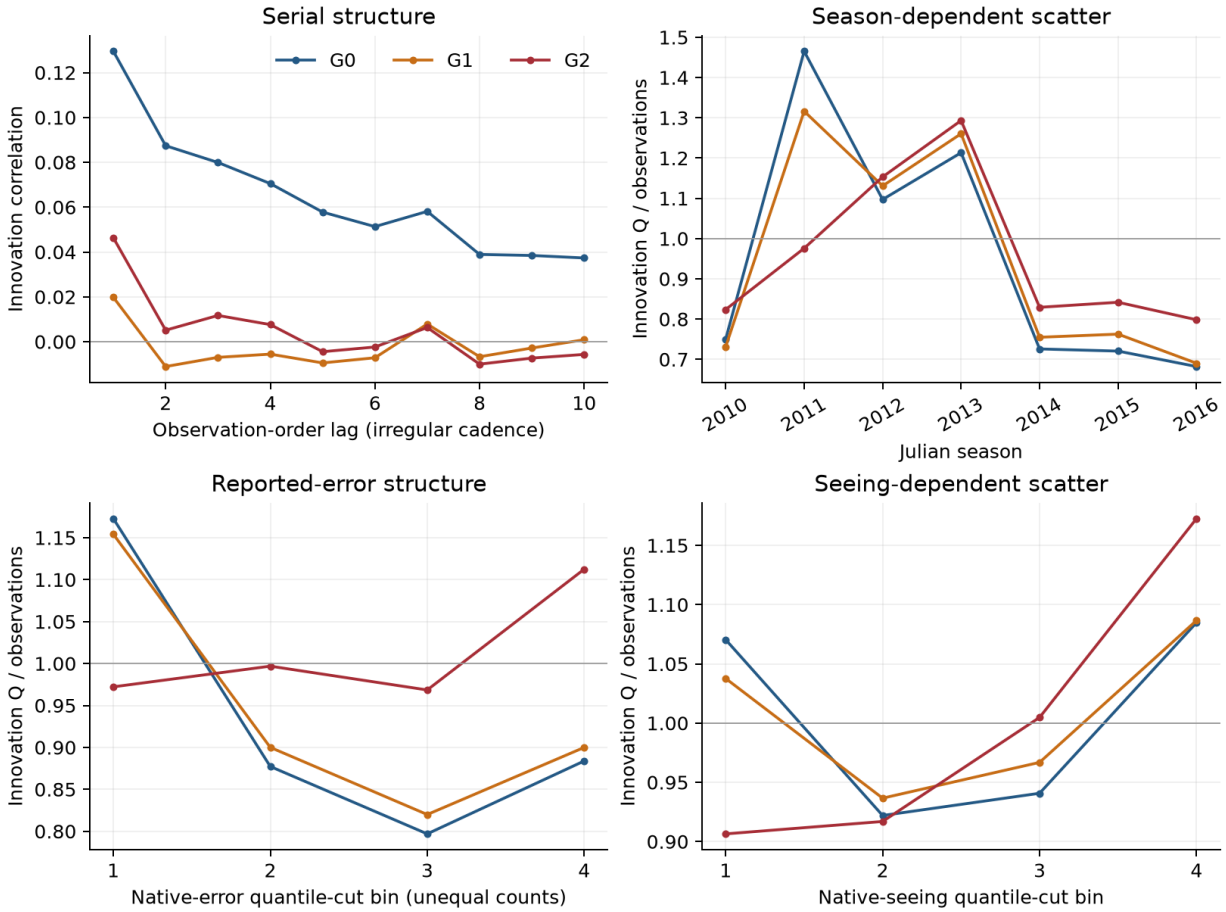


Figure 6: Residual structure for the three frozen joint candidates, after the declared continuation. Global quadratics conceal season, reported-error and seeing dependence. The reported-error quantile cuts have tied rounded values and unequal counts $(5817, 2874, 1787, 2022)$; they are not equal-sized samples. These diagnostic ratios are not calibrated confidence bands.

values. G1's measured held-out quadratic also exceeds all its generated values, despite narrowly passing the plug-in screen. These comparisons reinforce the limitations of a single scalar screen, but provide only the coarse 1/21 Monte Carlo tail resolution. They are local generating-mode checks, rather than global search or confidence-coverage certification. Ten G2 generating-mass profiles encounter the numerical timescale cap and remain interval-ineligible.

The completed serial mass grids retain 31 points per OU candidate, without cross-mass warm starts. G1 has 24 locally eligible and seven cap-limited rows; crossing refinement gives conditional support 5.989–8.399 $M_\odot$ at $\Delta D = 1$ and 5.021–9.910 $M_\odot$ at $\Delta D = 3.841459$. These are diagnostic likelihood supports under an inadequate noise hypothesis, not accepted confidence intervals. Every G2 grid row is cap-limited, and its 32-point refinement gives no eligible crossings or conditional support interval. An auxiliary parallel scheduling check preserves the serial results as authoritative: all compared admission states agree, but some optimizer values differ by up to 0.0133 in deviance across thread environments. Three separately repeated points with the same one-thread setting are identical. No cross-platform numerical identity or profile coverage is inferred from those checks.

The conditional angular scales range from 4.77 to 5.18 mas, whereas the conditional parallaxes range from 0.0324 to 0.0971 and the timescales from 282.4 to 340.5 d. The narrower variation in angular scale is a description of these fitted models, rather than a validated systematic error bound. In particular, their mass ratio is not stable across the fixed noise hypotheses. The software recovers conditional trajectories and angular signals, but this campaign does not identify a reliably calibrated mass or mass interval.

The protocol therefore stops mass acceptance for every fixed candidate. Diagnostic profiles and generated controls remain auditable; no additional covariance family is introduced to make an expected mass appear. Sensitivity calculations cannot convert an already inadequate baseline into accepted physical inference. The additional version 2 timing/frame/blend sensitivities are not executed, because baseline adequacy is their declared entry gate; no robustness to those systematics is claimed. Earlier version 1 diagnostic sensitivities remain separately preserved. The known-system analysis consequently provides no positive demonstration that this engine can reliably classify unknown black holes. It does demonstrate native-data ingestion, physical forward prediction, explicit failure detection, and replay.

For a previously unknown object, mass support above a declared compact-object threshold would still require independent darkness/lens-flux and single-lens compatibility evidence before a black-hole interpretation. Published darkness arguments for OB110462 [35] are separately attributed; this engine does not manufacture a new lens-flux limit or assume an absolute distance. The benchmark does not retrospectively turn the pilot's two archival follow-up light curves into mass measurements or new black-hole discoveries.

Useful next calibration evidence includes documented OGLE clock and per-exposure variance components, matched-star and seeing/detector diagnostics, and exposure-level HST astrometric covariances, reference transforms, bias corrections, and epoch weights. Absolute frame calibration is separately required before distance or absolute motion. A new unknown-system interpretation also needs measured angular-parallax information and independent lens/source flux constraints. A present-day image cannot manufacture the missing angular trajectory around a historical photometric peak. These are observational and calibration requirements, rather than permission to iterate the model until a black-hole claim passes.

Rejecting these covariance hypotheses does not prove that every estimator using the observations must fail. Correct-mean estimating equations with cluster/block robust inference are a separate possible route [38, 39]. They still need justified mean/exogeneity and between-block dependence assumptions, and finite-sample validation for the sparse, unevenly informative HST visits. Robust uncertainty does not correct a biased mean or shared frame calibration. No such alternative mass estimator is fitted or validated here, and no reliable mass support is inferred from the saved G0 point alone.

12 What could actually be tracked?

The resolved-pair product demonstrates compact-lens inference under its declared synthetic measurement and covariance model. The separate observational pilot demonstrates native-data ingestion, audited photometric screening and honest unavailable mass outputs. It identifies two historical archival-follow-up targets, but no mass-supported new black-hole interpretation or detectable BHSM envelope. The separate known-system joint benchmark also fails its declared noise-adequacy gates. Its conditional mass points do not establish a reliable known-black-hole measurement or reliability for unknown systems. The earlier broad information scan has a distinct limitation. For the illustrative cosmological mass and transverse speed 300 km s^{-1}, ten observer years move the center only 0.00236 pc, or $6.95 \times 10^{-6} r_E$. Traversing the broad synthetic $3.6 r_E$ scan at that speed would take about 5.2 million observer years. The short-span design has nearly dependent columns; its released spectrum must not be promoted into a realistic parameter forecast. Several independent source configurations could sample a broad spatial range, but that would require an actual source population, their redshifts and a corresponding data model. It is not automatically supplied by nine synthetic points.

A translated static profile is an adiabatic template evaluated at lens crossing time. It requires slow motion, control of coefficient evolution over the relevant crossing times, and control of scalar propagation. For the speed example $v/c \simeq 10^{-3}$, even nominal relative corrections at that level cannot be ignored when claiming the tiny subleading delay. Positive leading principal coefficients establish the local branch's hyperbolicity/ellipticity conditions; they do not prove global nonlinear stability, a UV completion or horizon admissibility. Fast intrinsic deformation or ringdown needs a causal metric solution. Such time-dependent black-hole lensing already has separate explicit models [40]; they do not close the R1 matching problem.

The unknown flux is especially consequential. The regular-body integrated source equation supplies one proxy, whereas a black-hole exterior has an integration constant that must be selected by horizon, interior and cosmological boundary conditions. The searched BHSM record did not establish that selection. A surface-acceleration identity or a Brown–York energy inversion on a known Schwarzschild worldtube does not determine this scalar flux or its deformation response. Nor can the two odd line-of-sight tensor modes be reconstructed by improving the precision of the same static projection.

This missing evaluation is specific to the present action and background. Cosmologically induced black-hole charges have been derived in related shift-symmetric Horndeski models [41], and regular stable quasi-stationary cubic-Galileon black holes with cosmological asymptotes have been constructed [42]. Those works provide a concrete matching strategy. The explicit self-accelerating cubic de Sitter constructions are not the present canonical-sign, linear-potential, rolling matter background; directly substituting the latter coefficients into that self-accelerating branch does not give a real Hubble parameter. Their results therefore cannot simply select the flux proxy used here, nor do they provide this paper's normalized quadrupole state.

Even perfect matching and a detectable transfer would track an exterior compact-mass/envelope combination. Any alternative interior producing the same Weyl response on every sampled ray gives the same data. A horizon claim therefore needs additional physics or independent observables. Unrestricted source transformations, line-of-sight potentials, uncertain distances, evolving macrostructure and flux propagation effects can also change the finite-family inference result. Multiple source redshifts alone do not remove every such freedom.

13 Discussion and conclusions: scientific advance and limits

Static cubic screening, its multipole powers and exact Green functions are established [7–10]. General STF gravitational multipole projection is also established [11]. The point-mass delay/flux

estimator is standard, and ordinary radial multipole separation and profiled linear inference are standard methods. We do not claim priority for those ingredients.

The addition combines their normalized R1 optical composition with an exterior-wide sufficient state cap and a runnable compact-lens inference workflow. Its theory exposes a delay-blind leading response, an exact apparent-mass/shear degeneracy and line-of-sight nulls. Its product retains full covariance and source freedom for resolved pairs, and reports historically versioned synthetic recovery with limited local-interval coverage checks. Its new unresolved route preserves native observations and carries every pilot outcome, including inadequate models and unavailable physical quantities, into the released table. The finite harmonic control makes a cause ambiguity exact rather than assigning every structured residual to BHSM.

The result supports a theoretical and computational lens-modeling methods paper with a usable BHSM-interpreted research tool and a bounded observational screening pilot. Ordinary mass/motion recovery remains useful when identified by suitable data; unresolved light-curve timescales alone do not supply it. Agreement with the Einstein-sector response is allowed; it does not uniquely select the underlying theory. It does not yet establish a detectable BHSM deformation, horizon classification or a decisive new astrophysical discovery. Further R1 prediction needs a normalized black-hole/homogeneous-background boundary map selecting exterior flux and deformation dynamics. Resolved-pair use requires calibrated positions, geometric fluxes, delays and an adequate environmental model. Unresolved mass inference requires independent angular-scale and parallax information, verified clocks/frames and a validated joint likelihood. The two historical follow-ups warrant archival and imaging feasibility checks; their ranking supplies no mass or BHSM-specific detection. No new telescope detection or journal submission is reported.

The completed joint candidate set stops without an accepted mass. Its apparently satisfactory global quadratics and whole-night predictive screens cannot override failed residual structure or a numerical covariance-timescale cap. Valuable further evidence includes per-exposure variance, seeing/detector diagnostics, astrometric covariance and reference-frame information. A separate robust estimating-equation route would need its own justified mean/dependence assumptions and finite-sample validation; this campaign does not evaluate it. A known system's published classification is separately attributed and does not certify this engine; one successful system would also be insufficient to establish population reliability.

Data and code availability. The formal reviewer repository is Berger–Hopf Standard Model, with the product under `products/blackhole_tracker/` and this paper under `papers/bhsm_blackhole_tracker_20261021/`. The integration branch is the real-data integration branch; the separate Manuscript-Generation repository is a development workspace. The rc4 resolved, native-photometry and joint product source is pinned to formal revision `9f1768a7fbac`. The final release receipt pins the completed source revision, wheel, manuscript PDF and CI outcomes. The theory/source reconciliation is independently pinned to formal revision `705d4c7e2f9a`. Earlier synthetic/theoretical projects and their receipts remain versioned historical material.

The original OGLE project supplies the native pilot observations [27]; the 2022 benchmark is Zenodo 6607578 version 1.0.0, declared CC BY 4.0 [28]. For this pilot and older benchmark, native source files remain external and unmodified; the distribution provides source URLs, required citations, hashes and fetch/replay instructions. Raw OGLE redistribution is not assumed from download access. The public `real_data_pilot_20261022/` directory contains the frozen protocol, source/kernel pins, compact all-event results, profiles and fit diagnostics, all 100 residual plots, historical benchmark results and separate literature annotations. Complete native rows and source-line identities are retained in external replay products. Source recovery and metadata gaps are documented with archived primary-source hashes. Theoretical replay uses `scripts/reproduce.py`; product validation and interface instructions are in the product README. The installed app's analysis export is replayable through `python -m bhtracker replay analysis.zip --output replayed`; the native route preserves its data contract, original source identity, BHSM interpreta-

tion and unavailable mass fields.

The separate `joint_known_system_20261022/` audit provides the 2023-data joint equations, fixed candidate protocol, partition correction, compact results and residual diagnostics, source-pack manifest and rights, independent computational reviews, and portable reproduction instructions. The source pack preserves the CC BY 4.0 author observations [35, 43] and separately attributed public HST header metadata. Its source version does not replace the frozen 2022 benchmark or 100-event pilot. Final release metadata pins the formal BHSM code revision; the development-workspace scientific freeze is separately recorded in the audit.

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A First variation of the delay–flux mass estimator

The point-mass delay relation is standard; see, for example, Muñoz et al. [14]. This appendix derives its first variation under a general local lens-potential perturbation, including the displacement of the images in the magnification calculation. The result is a thin-lens projection rule; it selects neither a horizon response nor a physical surface-state amplitude.

Use the reference point-mass Einstein angle and observed time scale

$$\theta_E^2 = \frac{4GM}{c^2}\frac{D_{ls}}{D_l D_s}, \qquad t_M = \frac{4GM}{c^3}(1+z_l),$$
$$\boldsymbol{x} = \boldsymbol{\theta}/\theta_E, \quad \boldsymbol{y} = \boldsymbol{\beta}/\theta_E. \tag{84}$$

Hold the physical source, body mass, distances, redshift and screen fixed while varying the dimensionless potential,

$$\tau_\epsilon(\boldsymbol{x};\boldsymbol{y}) = \tfrac{1}{2}|\boldsymbol{x}-\boldsymbol{y}|^2 - \ln r - \epsilon p(\boldsymbol{x}), \qquad r = |\boldsymbol{x}|. \tag{85}$$

For $u = |\boldsymbol{y}| > 0$, put $s = \sqrt{u^2+4}$. On the source axis the reference images are $x_+ = (u+s)/2 > 1$ and $x_- = -1/x_+ < 0$. Continue the positive-parity minimum and negative-parity saddle without

crossing a critical curve or reversing their flux/time ordering. With $R = |\mu_+|/|\mu_-| > 1$ and $\Delta t = t_M(\tau_- - \tau_+) > 0$,

$$\ln R_0 = 4\,\mathrm{asinh}(u/2), \qquad h(u) = \tfrac{1}{2}us + 2\,\mathrm{asinh}(u/2),$$
$$M_{\rm app} = \frac{c^3\Delta t}{4G(1+z_l)h[2\sinh(\frac{1}{4}\ln R)]}. \tag{86}$$

Indeed, the lens equation $x - u - 1/x = 0$ gives $\mu_+ = x_+^4/(x_+^4 - 1)$, $\mu_- = -1/(x_+^4 - 1)$ and $\tau_- - \tau_+ = \frac{1}{2}(x_+^2 - x_+^{-2}) + 2\ln x_+ = h(u)$. The distance factors in the physical time prefactor cancel against θ_E^2, so $M_{\rm app} = M$ for every reference alignment. The aligned Einstein-ring limit is excluded from the two-image estimator.

Let $\delta = \partial_\epsilon|_0$ and define

$$H_0(\boldsymbol{x}) = D^2 \ln r = \frac{I}{r^2} - \frac{2\boldsymbol{x}\boldsymbol{x}^T}{r^4}, \qquad A_i = I - H_0(\boldsymbol{x}_i). \tag{87}$$

Differentiating the stationary lens equation at fixed $\boldsymbol{y}$ yields

$$\delta\boldsymbol{x}_i = A_i^{-1}\nabla p(\boldsymbol{x}_i), \qquad \delta A_i = -D^2 p(\boldsymbol{x}_i) - DH_0(\boldsymbol{x}_i)[\delta\boldsymbol{x}_i],$$
$$\delta \ln|\mu_i| = -\,\mathrm{tr}(A_i^{-1}\delta A_i). \tag{88}$$

The evaluation-point term is essential. Explicitly,

$$DH_0(\boldsymbol{x})[\boldsymbol{v}] = -\frac{2(\boldsymbol{x}\cdot\boldsymbol{v})I}{r^4} - \frac{2(\boldsymbol{v}\boldsymbol{x}^T + \boldsymbol{x}\boldsymbol{v}^T)}{r^4} + \frac{8(\boldsymbol{x}\cdot\boldsymbol{v})\boldsymbol{x}\boldsymbol{x}^T}{r^6}. \tag{89}$$

Stationarity removes image displacements from the first arrival-time variation, giving $\delta(\tau_- - \tau_+) = p(\boldsymbol{x}_+) - p(\boldsymbol{x}_-)$. Since $h' = s$ and $(\ln R_0)' = 4/s$, the total estimator variation is

$$\boxed{\delta \ln M_{\rm app} = \frac{p(\boldsymbol{x}_+) - p(\boldsymbol{x}_-)}{h(u)} - \frac{s^2}{4h(u)}\left[\delta\ln|\mu_+| - \delta\ln|\mu_-|\right].} \tag{90}$$

Thus a pure alignment displacement cancels. A changing physical mass or redshift prefactor is a separate variation; it must not be hidden in an unspecified potential normalization.

For the quadrupole $p = q(r)\cos[2(\varphi - \varphi_q)]$, the two reference image angles differ by π. Write $c_2 = \cos[2(\varphi_s - \varphi_q)]$, $r_+ = (u+s)/2$, $r_- = 1/r_+$, $a_r = 1 + r^{-2}$ and $a_t = 1 - r^{-2}$. The radial image displacement is $\delta x_r = q' c_2/a_r$. Equation (88) then gives

$$\delta \ln|\mu_i| = c_2 \mathcal{F}_q(r_i), \tag{91}$$
$$\mathcal{F}_q(r) = \frac{q''}{a_r} + \frac{q'/r - 4q/r^2}{a_t} + \frac{2q'}{r^3 a_r}\left(\frac{1}{a_r} - \frac{1}{a_t}\right), \tag{92}$$
$$\delta \ln M_{\rm app} = c_2 \mathcal{K}_q(u), \tag{93}$$
$$\mathcal{K}_q(u) = \frac{q(r_+) - q(r_-)}{h(u)} - \frac{s^2}{4h(u)}\left[\mathcal{F}_q(r_+) - \mathcal{F}_q(r_-)\right]. \tag{94}$$

The final term of $\mathcal{F}_q$ is the radial evaluation-point contribution. Tangential displacements contribute off-diagonal terms whose trace vanishes in the reference radial/tangential eigenbasis. The physical action or worldtube source must supply q, q', q'' in the same lens units before this linear projection is applied.

The following closed forms provide nuisance comparisons and independent checks. Defining $K_* = s^3/[uh(u)]$, direct substitution yields

$$\mathcal{K}_{r^2} = K_*, \qquad \mathcal{K}_{r^{-2}} = (u^2+1)K_*, \qquad \mathcal{K}_1 = K_*. \tag{95}$$

Hence a compact projected quadrupole $Br^{-2}\cos[2(\varphi-\varphi_B)]$ contributes $B(u^2+1)K_*\cos[2(\varphi_s-\varphi_B)]$, while an external shear in the convention $(\gamma/2)r^2\cos[2(\varphi-\varphi_\gamma)]$ contributes $(\gamma/2)K_*\cos[2(\varphi_s-\varphi_\gamma)]$. At one alignment the amplitudes are degenerate; several alignments distinguish these two specified shapes only to the extent that the nuisance assumptions hold.

Linearity also gives a nontrivial blind direction,

$$\boxed{\mathcal{K}_{r^2-1}(u)=0 \qquad \text{for every } u>0.} \tag{96}$$

This is an exact *first-order* null of the statistic, not an exact finite-perturbation equivalence or a vanishing lens deformation. The potential $(r^2-1)\cos 2\varphi$ generally changes images, delay and flux ratio. It is a local smooth-potential counterexample on image annuli; whether a specified physical source can generate it requires its own source/boundary analysis. Different radial potentials therefore do not automatically give identifiable tracker signals. An inference must compare the evaluated $\mathcal{K}_q$ with the justified nuisance response space, including its conditioning, rather than compare potential labels.

These formulas concern quasistatic geometric optics and two noncritical continued primary images. They do not describe near-horizon strong-deflection images, new-image total flux, differential extinction or unmatched variable source emission. Exact symbolic identities, Cartesian total-Hessian checks and finite on-axis polynomial branch checks are supplied with the reproduction sources; they are internal consistency tests, not observations.

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